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OF TECHNOLOGY

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KTH Master Thesis : Methodology for Analyzing Structural Response to Blast-Type Rapid Dynamic Loading Using SOFiSTiK

FE-MODELING IN SOFiSTiK

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Sammanfattning

Detta examensarbete undersöker den strukturella responsen hos armerade betongelement utsatta för explosionsliknande belastningar med hjälp av avancerad finita elementmodeller. I ingenjörspraxis dimensioneras explosionsresistenta konstruktioner ofta med förenklade ekvivalenta statiska metoder, vilka inte alltid kan fånga det transienta beteendet hos konstruktioner som utsätts för kortvariga belastningar.

Syftet med detta arbete är att utvärdera skillnaderna mellan ekvivalenta statiska metoder och icke-linjära dynamiska tidsdomänanalyser för armerade betongkonstruktioner. Ett numeriskt modelleringsramverk utvecklades i finita elementmiljön SOFiSTiK. Explosionens tryck–tidsförlopp genererades med empiriska relationer från Kingery–Bulmash och UFC 3-340-02 och implementerades som tidsberoende ytbelastningar.

Modelleringsmetoden verifierades först genom jämförelser med publicerade numeriska och experimentella studier av armerade betongbalkar och plattor utsatta för explosioner. Metodiken tillämpades därefter på två ingenjörsmässiga fallstudier: explosionsresponsen hos en anaerob rötningsanläggning samt ett olycksscenario där en tung container tappas på en armerad betongplatta.

Resultaten visar att icke-linjära dynamiska analyser fångar den transienta interaktionen mellan lastutveckling och strukturell deformation mer realistiskt än förenklade statiska metoder. Även om dynamiska simuleringar ofta förutsäger större deformationer leder de i allmänhet till lägre inre krafter tack vare energidissipationsmekanismer. Detta beteende belyser begränsningarna hos ekvivalenta statiska metoder och visar potentialen hos icke-linjära dynamiska analyser för att ge mer realistiska och mindre konservativa strukturella bedömningar.

Studien presenterar även ett praktiskt ramverk för att implementera explosionslaster i SOFiSTiK och illustrerar fördelarna med dynamisk modellering för ingenjörsmässiga dimensioneringsproblem.

Abstract

This thesis investigates the structural response of reinforced concrete elements subjected to blast-type loading using advanced finite element modelling. In engineering practice, blast-resistant design is often performed using simplified equivalent static methods, which may not fully capture the transient behaviour of structures subjected to short-duration loads.

The objective of this work is to evaluate the differences between equivalent static approaches and nonlinear dynamic time-history analyses for reinforced concrete structures. A numerical modelling framework was developed in the SOFiSTiK finite element environment. Blast pressure-time histories were generated using empirical relations from the Kingery–Bulmash and UFC 3-340-02 formulations and implemented as time-dependent surface loads.

The modelling approach was first verified through comparison with published numerical and experimental studies involving reinforced concrete beams and slabs subjected to blast loading. The methodology was then applied to two engineering case studies: the blast response of an anaerobic digester structure and the accidental drop of a heavy container on a reinforced concrete slab.

The results show that nonlinear dynamic analyses capture the transient interaction between load evolution and structural deformation more realistically than simplified static approaches. While dynamic simulations often predict larger displacements, they generally lead to lower internal forces due to energy dissipation mechanisms. This behaviour highlights the limitations of equivalent static methods and demonstrates the potential of nonlinear dynamic analysis to provide more realistic and less conservative structural assessments.

The study provides a practical framework for implementing blast loading in SOFiSTiK and illustrates the benefits of dynamic modelling for engineering design applications.

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TABLE OF CONTENTS

ACKNOWLEDGEMENTS 4

TABLE OF CONTENTS 6

1 INTRODUCTION 7

1.1 Motivation and Context 7

1.2 Previous Work in the Field 7

1.3 Identified Research Gap and Objectives 9

2 THEORY OF BLAST LOADING 10

2.1 Explosion phenomena: detonation and deflagration 10

2.2 Blast wave propagation 11

2.3 Free-air burst and surface burst configurations 11

2.4 Scaled distance and empirical blast parameters (Kingery–Bulmash / UFC) 13

2.5 Pressure–time representation of blast loads 14

2.6 Incident, dynamic and reflected pressures on structures 14

2.7 Eurocode perspective on accidental explosions 15

2.8 Implications for finite element implementation in SOFiSTiK 16

3 NUMERICAL MODELLING AND VERIFICATION IN SOFISTIK 18

3.1 General Overview of SOFiSTiK 18

3.1.1 Geometry, elements, and boundary conditions 19

3.1.2 Reinforced concrete modelling within QUAD elements 20

3.1.3 Mesh Strategy and Load Resolution 20

3.1.4 Material and section definition (AQUA) 21

3.2 Definition of Blast Loading 24

3.2.1 Space–time separation of dynamic loading 24

3.2.2 Implementation in SOFiLOAD 24

3.2.3 Pressure–Time History 24

3.3 Nonlinear Dynamic Analysis Procedure 26
3.3.1 Equations of motion 26
3.3.2 Time integration and nonlinear solution 27
3.3.3 Static vs dynamic solution path 28
3.4 Numerical Stability and Good Modelling Practice 28
3.4.1 Time step selection 28
3.4.2 Load discretization and mesh dependency 28
3.4.3 Damping Assumptions 28
3.4.4 Nonlinear convergence and result extraction 29
3.5 Model Verification 29
3.5.1 Verification Methodology 29
3.5.2 Verification case 1 : Reinforced Concrete Beam 30
3.5.3 Verification case 2 : Reinforced Concrete Slab 33
4 ENGINEERING CASE STUDIES 36
4.1 Case Study 1: Blast Assessment of a Digester Structure 36
4.1.1 Project Background 36
4.1.2 Blast Loading Scenario 37
4.1.3 Finite Element Model Description 37
4.1.4 Non linear Static Analysis 38
4.1.5 Nonlinear Dynamic Analysis 39
4.1.6 Comparison of Results 39
4.2 Case Study 2: Slab subjected drop of a heavy container 41
4.2.1 Project Background 41
4.2.2 Finite Element Model Description 42
4.2.3 Loading Scenario 42
4.2.4 Comparison of Results 44
5 CONCLUSION AND FUTURE WORK 52
6 REFERENCES 54

1 INTRODUCTION

1.1 Motivation and Context

Accidental or intentional explosions represent extreme loading scenarios that may significantly affect the structural integrity of reinforced concrete constructions. In critical infrastructures such as nuclear facilities, industrial plants, and strategic buildings, the ability to assess structural performance under blast loading is essential to ensure safety, serviceability, and regulatory compliance.

In engineering practice, blast-resistant design must balance two competing objectives: maintaining high safety standards while avoiding unnecessary structural over-dimensioning. Within execution studies, particularly in the nuclear sector, structural engineers are occasionally required to evaluate the response of reinforced concrete elements subjected to explosion scenarios. These assessments are commonly performed using simplified equivalent static approaches derived from dynamic considerations. While such methods provide practical and conservative estimates, they may not always reflect the actual transient structural behaviour.

Advances in computational mechanics have enabled the use of nonlinear dynamic time-history analyses to simulate the response of structures subjected to short-duration extreme loads. These methods allow the consideration of geometric nonlinearity and transient load application, offering a more detailed representation of structural behaviour during blast events. However, their implementation within conventional civil engineering workflows remains limited, and their practical advantages over simplified design approaches are not always clearly quantified.

In this context, there is a need to investigate the applicability and added value of nonlinear dynamic analysis for reinforced concrete structures subjected to blast loading within an engineering design environment. Establishing a consistent modelling framework and evaluating the differences between simplified and advanced approaches may contribute to a more rational and optimized structural design process.

1.2 Previous Work in the Field

The response of reinforced concrete (RC) structures subjected to blast loading has been extensively investigated over the past decades through analytical, experimental, and numerical approaches. The characterization of blast loads is commonly based on empirical relationships derived from large-scale detonation tests. Among the most widely used references are the formulations by Kingery and Bulmash [1], which provide airblast parameters as a function of scaled distance (see Section 2.4), and the Unified Facilities Criteria UFC 3-340-02 [2], which remains a primary design guideline for structures required to resist accidental explosions. These documents establish pressure–time relationships, impulse calculations, and scaling laws that form the basis for both simplified and advanced structural analyses.

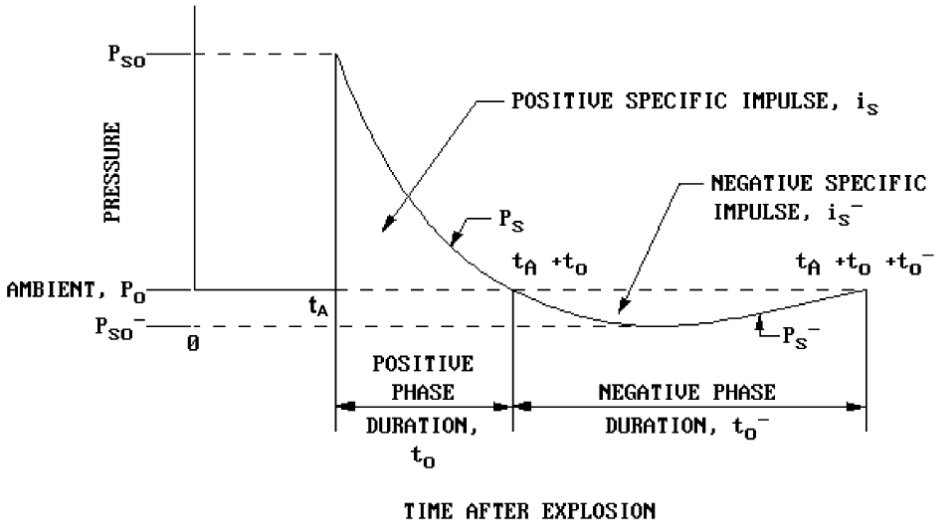


Figure 1. Idealized blast pressure–time history described by the Friedlander equation (adapted from UFC 3-340-02).

Experimental investigations have played a crucial role in improving the understanding of the behaviour of reinforced concrete (RC) elements subjected to blast loading. Numerous studies have investigated the response of individual structural components such as slabs, beams, columns, and wall panels. Among the many studies available in the literature, several representative examples can be mentioned. For instance, Talaat et al. [3] investigated the response of reinforced concrete slabs subjected to blast loads and compared experimental observations with finite element simulations, highlighting the influence of boundary conditions and reinforcement detailing on failure modes. Similarly, Zhang et al. [4] performed experimental investigations on reinforced concrete beams subjected to close-in blast loading, examining scaling effects and dynamic response characteristics. Their results showed that geometric scaling can significantly influence damage patterns and deformation capacity.

In addition to free-field explosive tests, controlled laboratory investigations using shock tubes have gained increasing attention. Shock-tube experiments allow the generation of repeatable pressure–time histories representative of blast waves while offering improved experimental control and safety. Among recent contributions, Rosin et al. [5] investigated reinforced concrete wall panels subjected to shock-tube loading, combining experimental measurements with numerical simulations to evaluate blast mitigation measures. Such studies provide valuable validation data for computational models and contribute to a better understanding of dynamic failure mechanisms in RC elements.

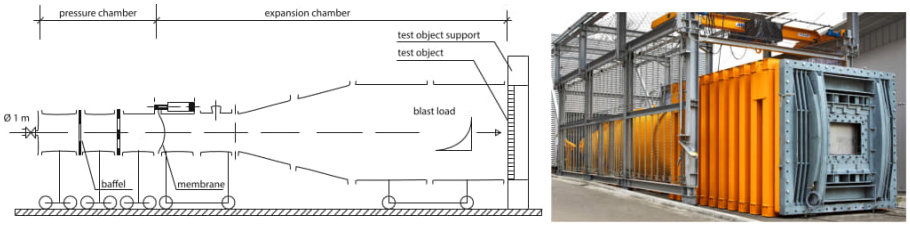


Figure 2. Shock-tube experimental setup used to generate controlled blast-type pressure waves for testing reinforced concrete wall panels (adapted from Rosin et al. [5]).

Parallel to experimental research, finite element (FE) modelling has become a domi-

nant tool for analysing structural response under blast loading. Advanced numerical simulations enable the representation of geometric nonlinearity, material behaviour, and transient dynamic effects. Many studies employ explicit dynamic solvers to capture high strain-rate phenomena and progressive damage in reinforced concrete members. Numerical models are often validated against experimental results to assess their predictive capabilities. Comparisons between measured and simulated responses generally show good agreement in terms of global deformation and failure modes, although discrepancies may arise due to uncertainties in material modelling and boundary conditions.

Overall, previous research demonstrates that the blast response of reinforced concrete elements depends on several interacting parameters, including scaled distance, structural geometry, reinforcement ratio, boundary conditions, and load duration. While substantial progress has been made in experimental characterization and numerical modelling of individual RC components, variations in modelling assumptions and analysis approaches continue to influence predicted structural performance.

1.3 Identified Research Gap and Objectives

Despite extensive experimental and numerical investigations on reinforced concrete structures subjected to blast loading, important gaps remain between research practices and engineering applications.

Equivalent static methods are widely employed in design studies due to their simplicity and ease of implementation. However, the degree to which these simplified approaches accurately represent the transient nonlinear response of reinforced concrete structures under blast loading remains uncertain. In particular, systematic comparisons between equivalent static analyses and nonlinear dynamic time-history simulations within a consistent modelling framework are limited.

Furthermore, while advanced blast simulations are frequently conducted using specialized explicit dynamic solvers, the implementation of such analyses within structural engineering software commonly used in execution studies remains less documented.

The present study therefore aims to:

- implement and apply an equivalent static approach to a representative reinforced concrete structural element;
- develop and perform a nonlinear dynamic time-history analysis of the same configuration;
- compare the predicted structural responses obtained from both approaches;
- and establish a practical computational framework within the SOFiSTiK software environment for engineering applications.

Through this comparative investigation, the study seeks to assess the potential of nonlinear dynamic analysis to provide a more refined yet practically applicable alternative to conventional equivalent static methods in blast-resistant structural design.

2.1 Explosion phenomena: detonation and deflagration

Explosions correspond to a rapid release of energy that generates a pressure wave propagating through the surrounding medium. Depending on the physical mechanism governing the combustion process, two main types of explosive phenomena are commonly distinguished: *detonation* and *deflagration*.

A **detonation** is characterized by a supersonic reaction front coupled with a shock wave propagating through the explosive material. The chemical reaction occurs almost instantaneously behind the shock front, producing extremely high pressures and temperatures. High explosives such as TNT or RDX typically detonate and generate blast waves with very steep pressure rises and short durations.

In contrast, a **deflagration** corresponds to a subsonic combustion process driven primarily by heat transfer and mass diffusion. The reaction front propagates through the combustible mixture at a velocity lower than the speed of sound in the medium. Deflagrations typically occur in accidental gas explosions where the combustion of a fuel–air mixture leads to a progressive pressure build-up. For a typical pressure-time history associated with a detonation, see Figure 4.

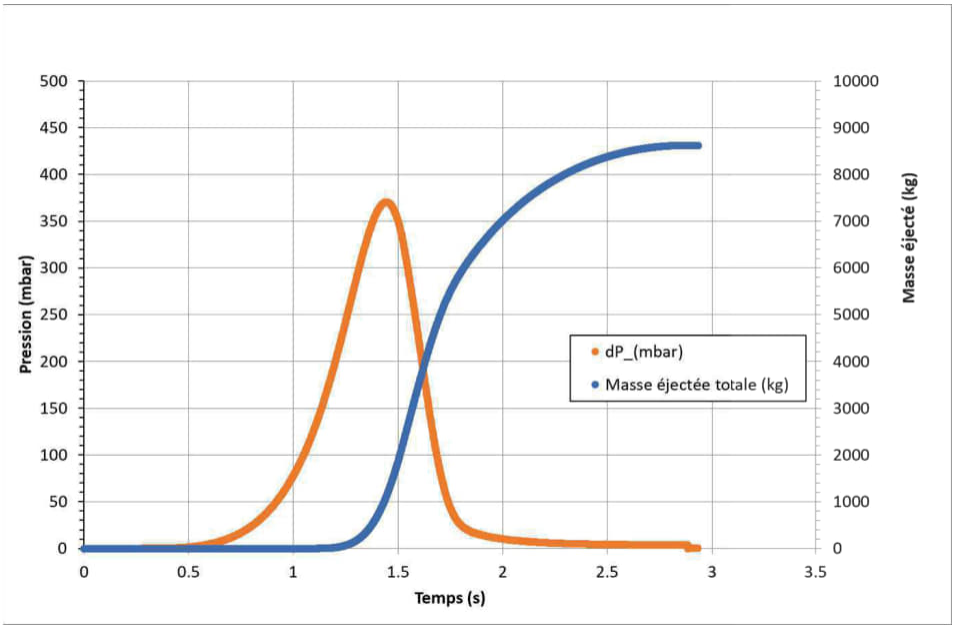


Figure 3. Typical overpressure-time history associated with a deflagration event (orange curve) used as a case study in this report. Compared with detonation phenomena, deflagrations generally exhibit a slower pressure rise and a longer duration pressure load.

Because of these different propagation mechanisms, the resulting pressure-time histories differ significantly. Detonations generally produce extremely rapid pressure peaks associated with strong shock waves, whereas deflagrations generate slower pressure rise times and longer duration pressure loads.

The Eurocode framework (EN 1991-1-7, Annex D) primarily addresses accidental gas explosions developing as deflagrations, whereas blast-resistant design guidelines such as UFC 3-340-02 focus mainly on external blast waves generated by high-explosive detona-

tions.

2.2 Blast wave propagation

When a high explosive detonates in air, the rapid release of energy generates a shock front that propagates radially outward from the explosion centre. This shock wave is characterized by a sudden increase in pressure, density, and temperature across the shock front.

As the wave propagates away from the explosion source, its intensity gradually decreases due to geometric spreading and energy dissipation in the surrounding medium. The pressure variation experienced at a given location therefore consists of a sharp pressure rise followed by a gradual decay toward ambient atmospheric pressure.

This transient pressure evolution is typically described by a pressure–time history composed of a positive phase and, in many cases, a subsequent negative phase.

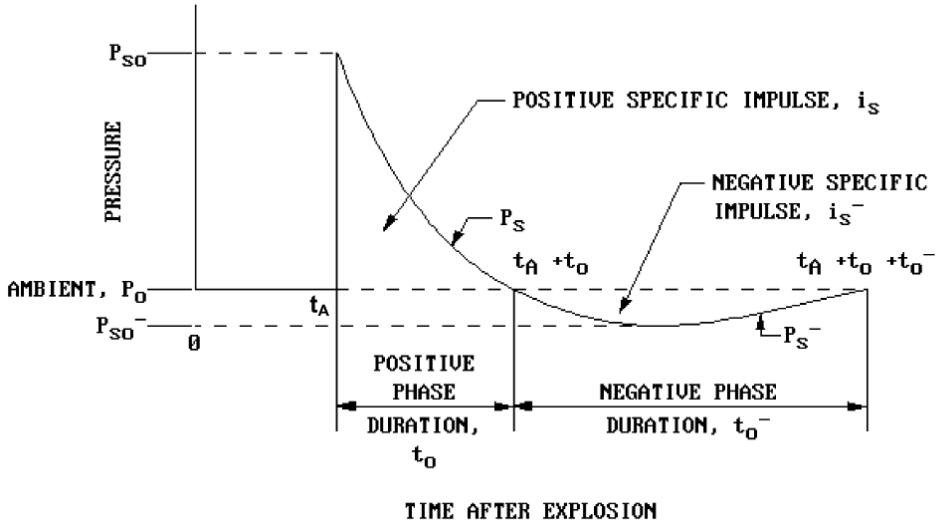


Figure 4. Typical idealized blast pressure–time history illustrating the positive and negative phases of a free-field blast wave.

The positive phase corresponds to the period during which the pressure remains above ambient pressure and generally governs the peak structural response. The negative phase represents a suction effect following the passage of the shock front.

For many reinforced concrete structures subjected to short-duration impulsive loading, the influence of the negative phase on the global structural response is relatively limited. Consequently, although it may be included for completeness in numerical simulations, it is often neglected in simplified design approaches.

2.3 Free-air burst and surface burst configurations

The characteristics of blast loading depend strongly on the location of the detonation relative to the ground surface. Two idealized explosion configurations are commonly distinguished in blast engineering.

- **Free-air burst:** the explosive charge detonates above the ground surface and the

blast wave initially propagates spherically in air before interacting with the ground and surrounding structures.

- **Surface burst:** the explosive charge detonates at or very near the ground surface, producing a hemispherical shock wave and a stronger near-ground pressure environment.

This distinction is essential because empirical blast datasets, such as the Kingery–Bulmash relationships, are derived for specific detonation configurations. Blast parameters must therefore be selected consistently with the explosion scenario under consideration.

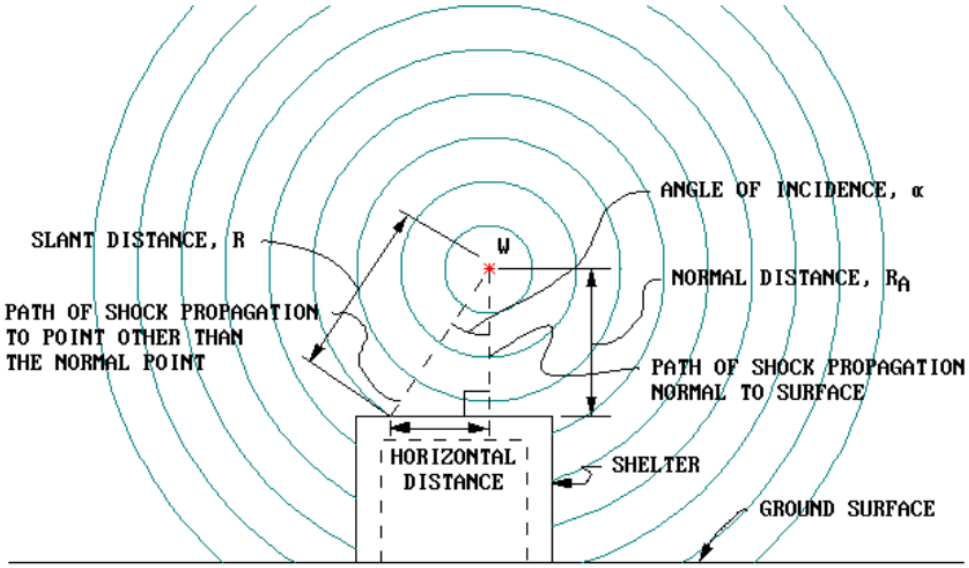


Figure 5. Free-air burst blast environment and geometric definitions including the slant distance R and the angle of incidence α.

When the burst occurs sufficiently close to the ground surface, the incident shock wave interacts with the ground-reflected wave. Under certain conditions these waves may merge and form a **Mach stem**. The point where the incident wave, reflected wave, and Mach front intersect is known as the triple point.

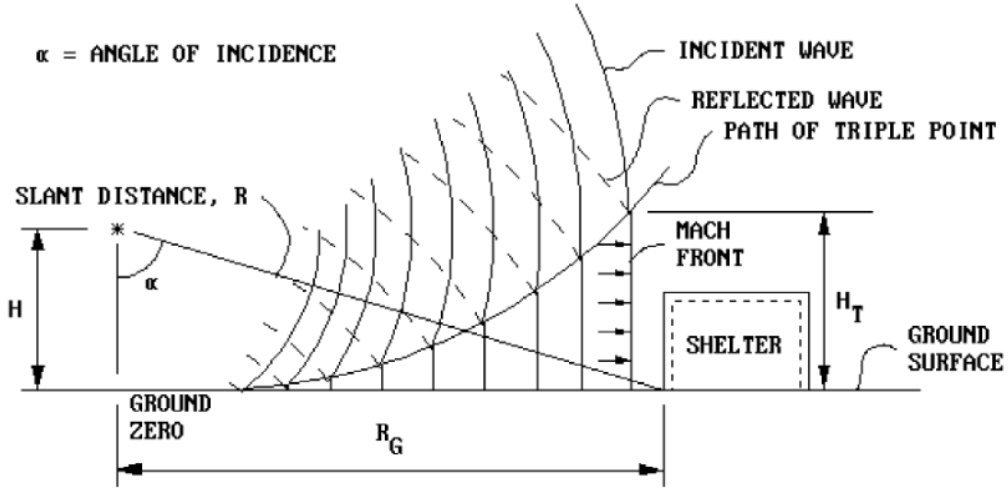


Figure 6. Air-burst environment near the ground showing incident and reflected waves and the formation of a Mach stem beyond the triple point.

The presence of Mach reflection modifies both the magnitude and the temporal distribution of the pressures acting on structures.

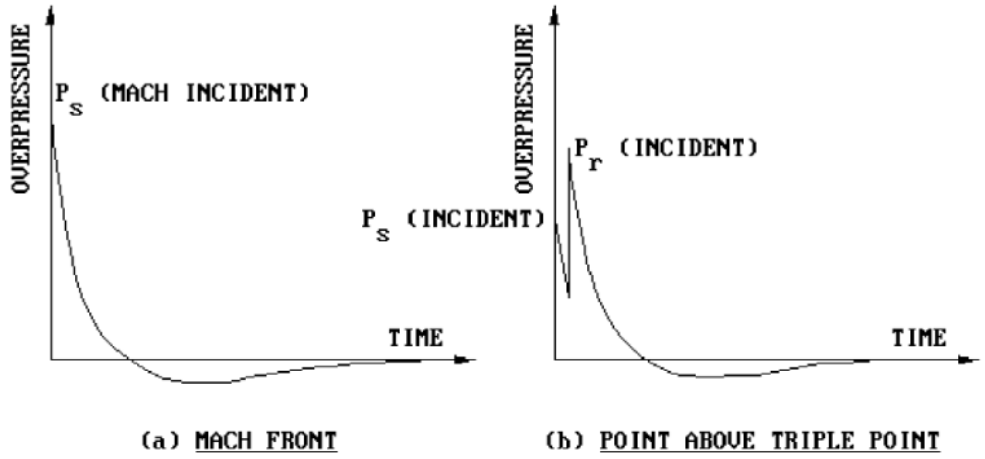


Figure 7. Qualitative pressure–time histories associated with Mach reflection conditions.

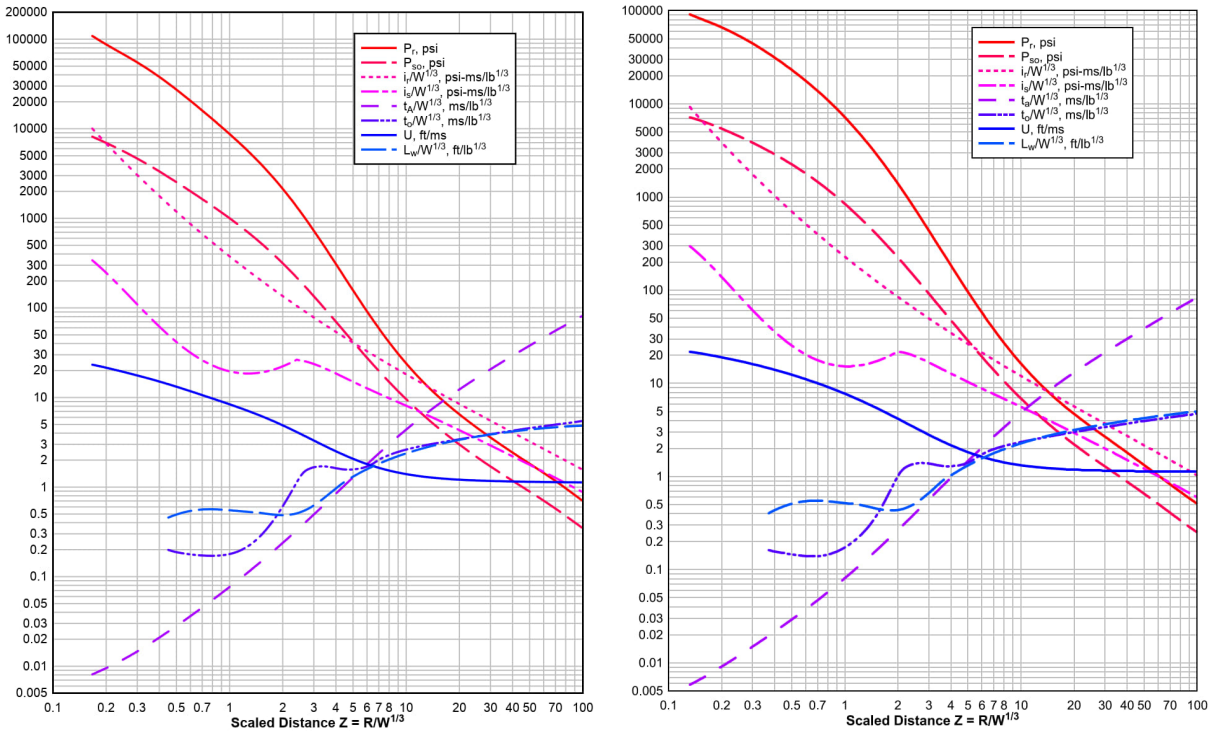
2.4 Scaled distance and empirical blast parameters (Kingery–Bulmash / UFC)

Blast effects depend primarily on the explosive charge weight and the distance from the explosion. In order to compare explosions of different magnitudes, blast engineering commonly relies on the Hopkinson–Cranz scaling law, which introduces the scaled distance Z:

$$Z = \frac{R}{W^{1/3}}$$

where R represents the stand-off distance and W is the TNT equivalent charge mass.

Using this normalization, empirical relationships derived from the Kingery–Bulmash database express key blast parameters such as peak incident overpressure P_{so}, positive phase duration t_0^+, and positive impulse i_s^+ as functions of the scaled distance.



(a) Hemispherical surface burst

(b) Spherical free-air burst

Figure 8. Kingery–Bulmash empirical blast parameters expressed as functions of the scaled distance for hemispherical surface bursts and spherical free-air bursts.

2.5 Pressure–time representation of blast loads

For numerical structural analyses, the discrete blast parameters obtained from empirical relationships must be converted into a continuous pressure–time function. The positive phase of the blast wave is commonly represented using the Friedlander equation

$$P(t) = P_{so}\left(1 - \frac{t}{t_0^+}\right)\exp\left(-b\frac{t}{t_0^+}\right)$$

where b is a decay coefficient controlling the exponential decay of the pressure.

The associated positive specific impulse is defined as

$$I^+ = \int_0^{t_0^+} P(t)dt$$

and represents the total momentum transferred during the positive phase of the blast loading.

2.6 Incident, dynamic and reflected pressures on structures

The pressure acting on a structural surface is not always equal to the free-field incident overpressure alone. UFC distinguishes three main pressure components:

- **Incident pressure** associated with the propagation of the shock wave in the free field.

- **Dynamic pressure** associated with the momentum of the air particles behind the shock front.
- **Reflected pressure** resulting from the reflection of the shock wave on the structural surface.

When a shock wave impinges on a rigid surface, the reflected pressure may significantly exceed the incident pressure.

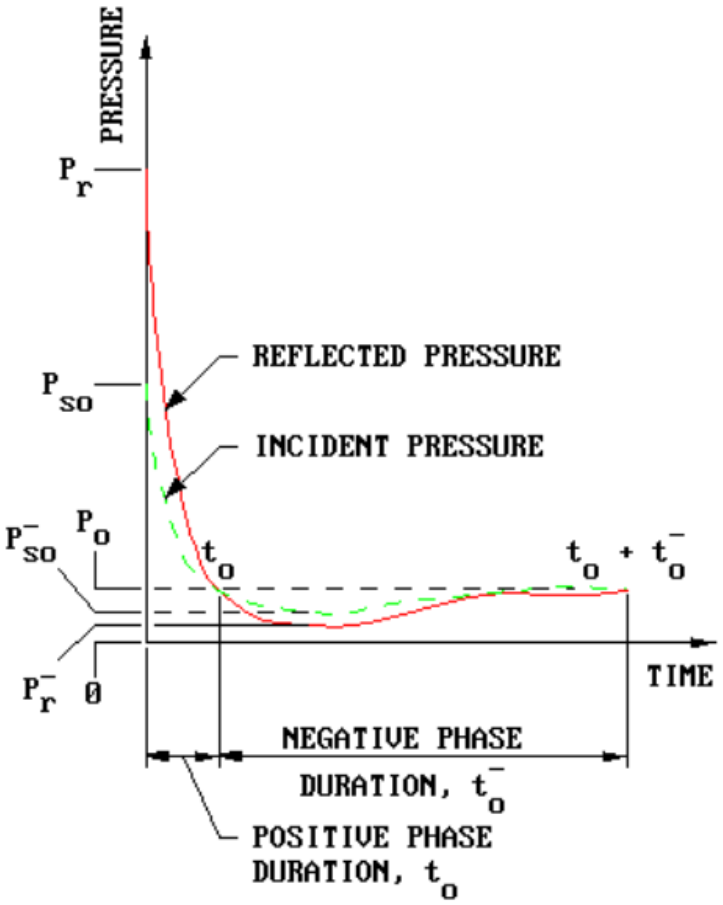


Figure 9. Comparison between incident and reflected pressure–time histories for a free-air burst.

2.7 Eurocode perspective on accidental explosions

Within the Eurocode framework, accidental actions are addressed in *EN 1991-1-7 (Eurocode 1 – Actions on Structures – Part 1-7: Accidental Actions)* [6].However, the treatment of explosions remains limited compared to dedicated blast design standards. Explosion-related provisions are provided in Annex D (informative), which offers guidance primarily for internal gas explosions in buildings.

As an informative annex, Annex D does not establish mandatory design rules, but rather presents simplified approaches applicable to specific scenarios. The focus is placed on gas explosions resulting from deflagration processes, the estimation of maximum internal overpressure, and the design of venting areas to mitigate structural damage. The objective is therefore pressure relief and damage limitation, rather than detailed modelling of external blast waves.

2.8 Implications for finite element implementation in SOFiSTiK

The determination of blast parameters plays a central role in the numerical implementation of blast loading, particularly when using software such as SOFiSTiK. In explicit finite element solvers such as LS-DYNA or Abaqus, integrated blast-loading modules allow the user to define the explosive scenario through a limited set of parameters, typically the TNT equivalent mass and the stand-off distance. Based on these inputs, the software automatically evaluates the blast wave characteristics and generates the corresponding pressure–time history applied to the structure. An example of this approach in LS-DYNA is illustrated in Fig. 10, where the explosive charge is defined at a given distance from the structure.

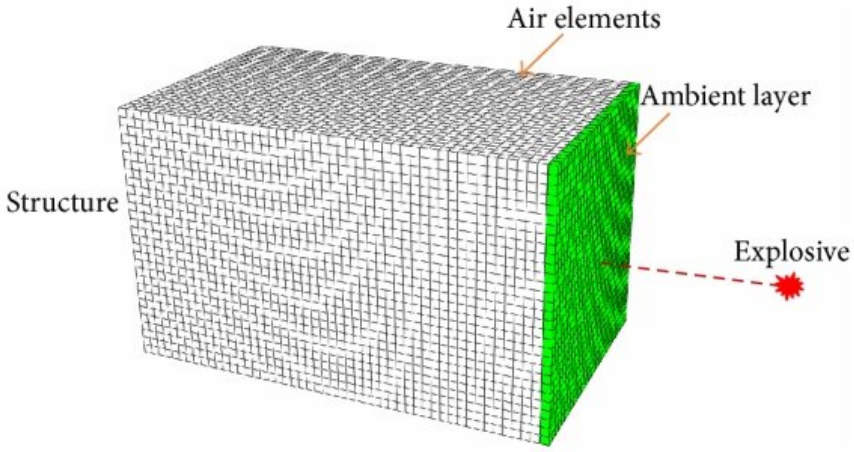


Figure 10. Example of blast modelling in LS-DYNA where the explosive charge is defined by its TNT equivalent mass and its stand-off distance from the structure.(adapted from [7]).

By contrast, SOFiSTiK does not provide a direct built-in blast generator. The analyst must therefore determine the relevant blast parameters explicitly and use them to construct the pressure–time function representing the blast wave, for example using a Friedlander-type curve or a simplified triangular representation. This function is then applied as a time-dependent load acting on the exposed structural surfaces in the transient dynamic analysis. The implementation of such a load definition in the SOFiPLUS environment is illustrated in Fig. 11.

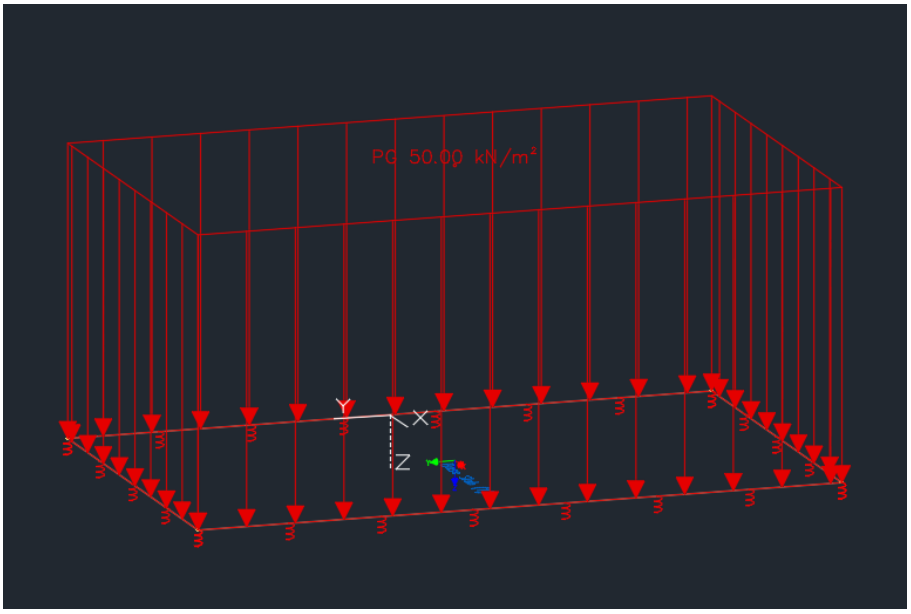


Figure 11. Example of blast load definition in SOFiSTiK through the SOFiPLUS interface, where the pressure is directly applied to the structural surface.

Consequently, when using SOFiSTiK, the blast parameters are not only theoretical descriptors of the explosion but also the quantities that directly define the pressure–time loading function applied to the structure.

3 NUMERICAL MODELLING AND VERIFICATION IN SOFISTIK

3.1 General Overview of SOFiSTiK

SOFiSTiK is a finite element (FE) structural analysis software primarily aimed at building and civil engineering applications. Its workflow is organized around a modular architecture, in which specialized programs exchange information through a common central database. This modular structure is particularly well suited to nonlinear dynamic problems, as it allows a clear separation between: (i) material and section definition, (ii) load modelling, and (iii) static and dynamic solvers.

In this thesis, the modelling and analysis chain relies mainly on the following modules:

- **AQUA** for material and cross-section definition (reinforced concrete, steel, etc.) [8].
- **SOFiLOAD** for the definition and management of load cases and time-dependent load functions [9].
- **ASE** for static and dynamic analyses, both linear and nonlinear, using implicit time integration methods [10].
- **DYNA** for eigenvalue analysis and dynamic simulations, linear or nonlinear, using explicit time integration methods [11].
- **TEDDY** as the text-based input interface used to define and control the computational workflow. It allows the user to write command scripts directly in SOFiSTiK's input language, making it possible to implement customized procedures and mathematical functions. In the context of this work, TEDDY is particularly important for defining the pressure–time functions associated with blast loading, enabling the explicit implementation of the overpressure curve used in the transient dynamic analysis.

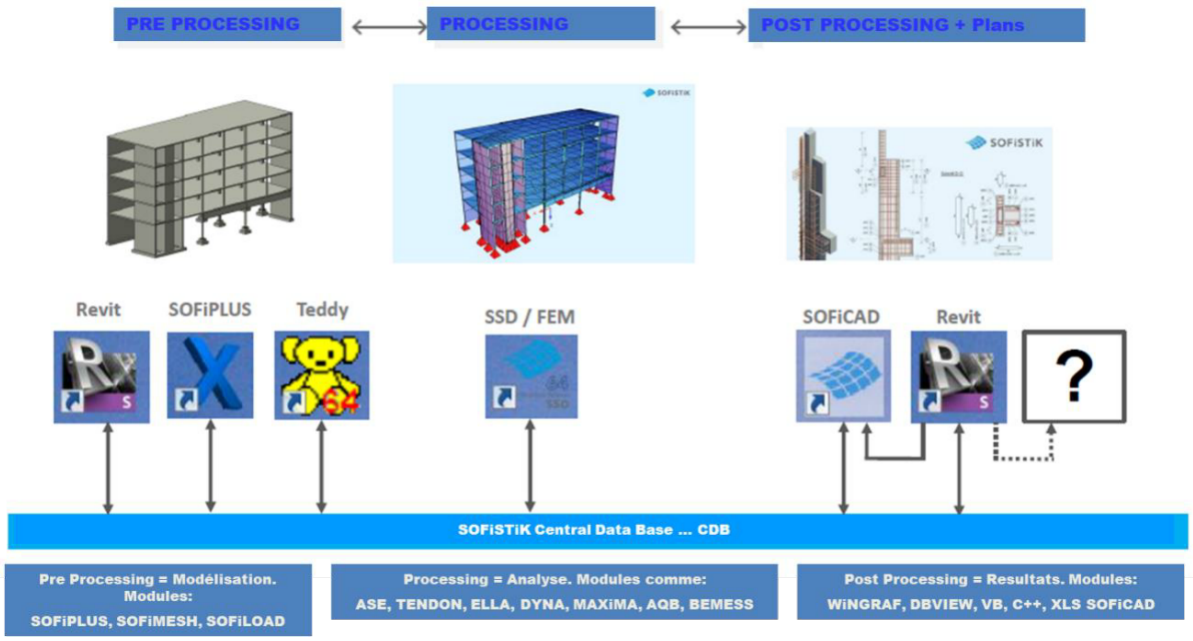


Figure 12. Typical workflow adopted in this study: material definition (AQUA), load modelling (SOFiLOAD), nonlinear static analysis (ASE), and nonlinear dynamic analysis (DYNA).

The overall computational workflow adopted in this study is illustrated in Figure 13. It summarizes the main modelling stages, from the definition of blast parameters to the comparison between nonlinear static and nonlinear dynamic analyses.

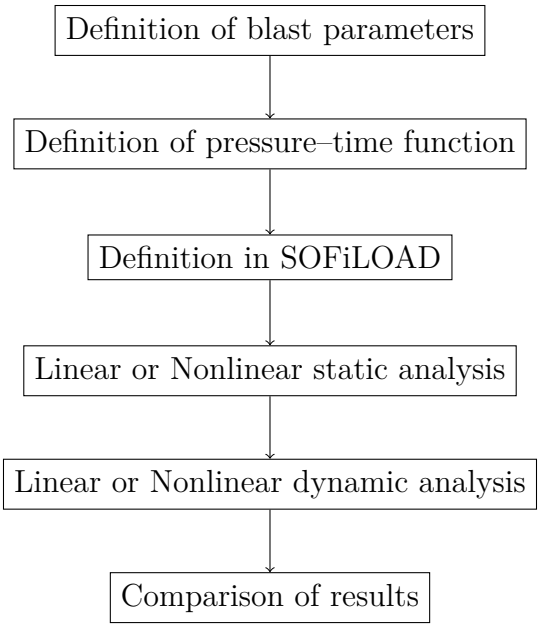


Figure 13. Computational workflow used in this study for the implementation of blast loading and the comparison between nonlinear static and dynamic analyses.

3.1.1 Geometry, elements, and boundary conditions

The reinforced concrete panels are discretized using four-node quadrilateral finite elements (QUAD), which represent general plane elements capable of modelling membrane, bending and shear behaviour simultaneously. These elements are based on a combined formu-

lation in which in-plane (membrane) actions and out-of-plane (plate bending) effects are coupled, allowing an accurate representation of plate and shell responses **SOFiMSHA**. Each QUAD element possesses four nodes and can degenerate into a triangular element if required

A local coordinate system is associated with each element, with the out-of-plane direction defined normal to the element surface, while in-plane directions govern membrane and shear responses. Internal forces (membrane forces, bending moments and shear forces) are evaluated within this local system.

This formulation provides a good compromise between computational efficiency and accuracy, making QUAD elements particularly well-suited for modelling reinforced concrete slabs subjected to dynamic loading such as blast effects.

The structural systems studied are discretized using QUAD elements consistent with the target behaviour (global bending response and local stress distribution). Boundary conditions are defined to represent the actual support conditions of each case study, ensuring that the resulting stiffness and modal properties are realistic. Particular attention is paid to avoiding artificial constraints that may distort the dynamic response (e.g., over-constraining rotational degrees of freedom).

3.1.2 Reinforced concrete modelling within QUAD elements

In non-linear analyses, reinforced concrete is modelled as a heterogeneous system in which concrete and reinforcement contribute differently to the internal force equilibrium.

The QUAD elements provide membrane forces and bending moments in their local coordinate system, which are subsequently used for the reinforced concrete formulation. The reinforcement is defined in terms of equivalent steel areas (in cm^2/m) for each direction and layer (e.g., top and bottom), using the BEMESS module. These reinforcement layers are aligned with the local coordinate system of the finite elements.

The internal forces obtained from the finite element analysis (normal forces and bending moments) are transformed into forces acting in the reinforcement directions. Due to the limited tensile strength of concrete, tensile forces are primarily carried by the reinforcement, while compressive stresses are mainly resisted by the concrete. However, reinforcement may also contribute to compression when required by the stress state. In particular, shear and skew stress states are resisted by the interaction between the reinforcement mesh and compressive struts developing in the concrete.

This approach allows the modelling of reinforced concrete as a composite system in which the load transfer is governed by the compatibility of strains and the equilibrium of internal forces, rather than by assuming an equivalent homogeneous material. Such a formulation is particularly suitable for non-linear analyses, where cracking, redistribution of stresses and directional effects of reinforcement play a significant role.

3.1.3 Mesh Strategy and Load Resolution

When blast pressure is applied as a distributed surface load, the finite element mesh directly influences the numerical representation of the load. A mesh that is too coarse

may lead to inaccurate load distribution and stress evaluation, while an excessively refined mesh increases computational cost without significant improvement in global response.

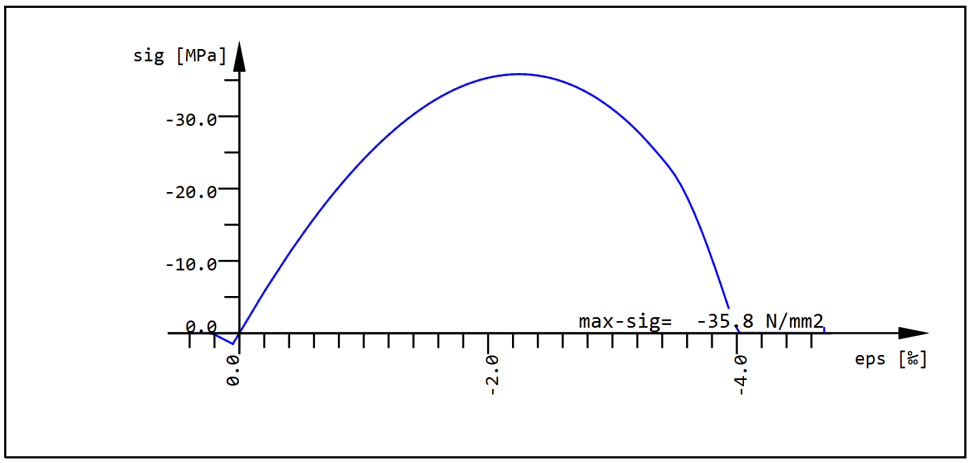
3.1.4 Material and section definition (AQUA)

Material properties and cross-sections are defined using the AQUA module of SOFiSTiK [8]. In reinforced concrete modelling, the definition of constitutive laws is critical to capture stiffness degradation due to cracking, nonlinear compressive behaviour, and the inelastic response of reinforcement. In the present study, the constitutive parameters are implemented consistently for both the nonlinear static (ASE) and nonlinear dynamic (DYNA) analyses in order to isolate the influence of inertia and time integration effects from purely material modelling differences.

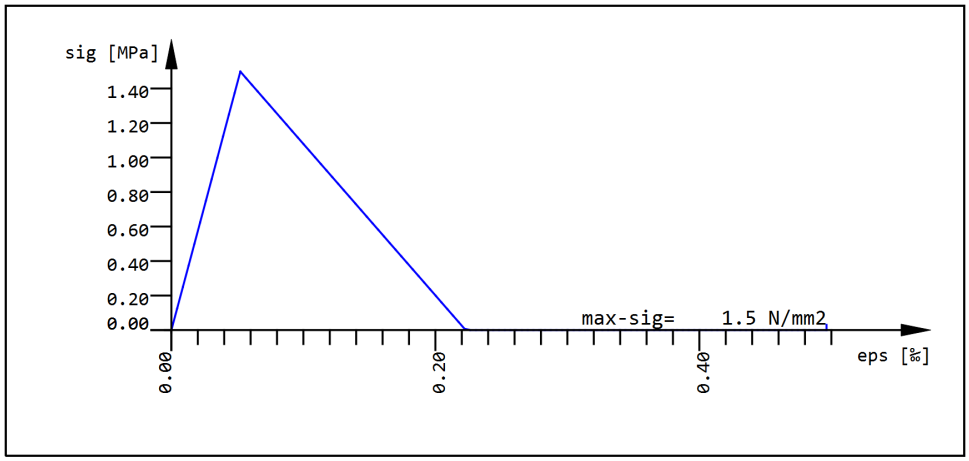
Figure 14 presents the TEDDY input used to define the material model in AQUA. The resulting concrete stress–strain relationship (including compression nonlinearity and tensile cracking assumptions) is shown in Figures 15a and 15b, while the reinforcing steel constitutive response (elastic–plastic with yielding) is illustrated in Figure 15c. These definitions provide a unified material basis for the comparative assessment carried out in Sections 3.5 and 4.

```
PROG AQUA urs:30.1
HEAD Slab impact
UNIT 5
NORM 'NF' 'en199x-200x' UNIT 5
CONC 10 C 35
SSLA CALC 1.2
SSLA    EPS          SIG          TYPE
        0.00         0.00         POL
       -0.20        -6.880969397  POL
       -0.40       -13.23143052   POL
       -0.60       -19.03203171   SPL
       -0.80       -24.26246852   SPL
       -1.00       -28.90142435   SPL
       -1.20       -32.92650656   SPL
       -1.40       -36.31417776   SPL
       -1.60       -39.03968163   SPL
       -1.80       -41.07696308   SPL
       -2.00       -42.39858184   SPL
       -2.20       -42.97561924   SPL
       -2.40       -42.77757729   SPL
       -2.60       -41.77226931   SPL
       -2.80       -39.92570149   SPL
       -3.00       -37.20194418   SPL
       -3.20       -33.56299218   SPL
       -3.40       -28.96861260   SPL
       -3.50       -26.29991510   SPL
       -4.00         0.00         POL
STEE 20    B 500B
SSLA CALC 1.0
SSLA    EPS          SIG          TYPE
        -50          -540         POL
       -2.17         -500         POL
          2.17       500             POL
           50     540                 POL
END
```

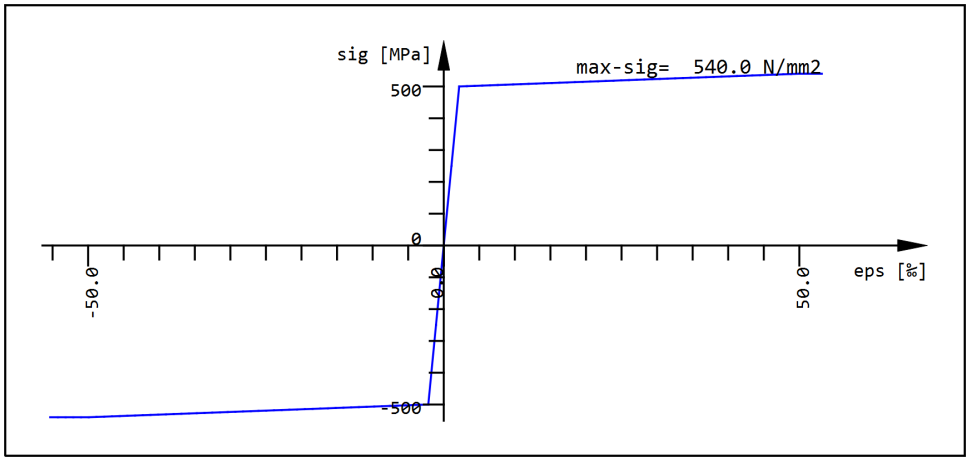
Figure 14. Exemple of TEDDY script defining the concrete material model in the AQUA module



(a) Concrete compressive stress–strain relationship used in the model.



(b) Concrete tensile stress–strain relationship used in the model.



(c) Steel stress–strain relationship used in the model.

Figure 15. Material constitutive laws implemented in the numerical model.

3.2 Definition of Blast Loading

3.2.1 Space–time separation of dynamic loading

In DYNA, time-dependent loading can be expressed as a superposition of multiple contributions, each defined by a spatial load pattern multiplied by a time function [11]. This separation is particularly convenient for blast modelling, where the spatial distribution may be taken as uniform over a loaded surface while the time variation follows a prescribed pressure–time history. The load vector can be written as:

$$\mathbf{p}(t) = \sum_{j=1}^{n_p} \mathbf{p}_{0,j}\, g_j(t), \tag{1}$$

where $\mathbf{p}_{0,j}$ is the spatial load pattern (constant in time) and $g_j(t)$ is the associated time function [11].

3.2.2 Implementation in SOFiLOAD

Blast loading is applied as a pressure acting on structural surfaces, defined in SOFiLOAD as element or area loading and associated with a dedicated load case and action [9]. SOFiLOAD handles the transition from geometric load definitions to FE-consistent nodal/element loads; hence, the spatial discretization must be adequate for representing the pressure distribution [9]. For transient analysis, the pressure history is introduced through time functions, which can be defined directly or imported, and then assigned to the relevant load case.

3.2.3 Pressure–Time History

The blast load is introduced in the numerical model through an idealized pressure–time function applied to the exposed structural surface. The time-dependent load is defined in SOFiLOAD and associated with the dynamic analysis module (DYNA), following the space–time separation described previously.

An excerpt of the implemented load-time definition in TEDDY is presented in Figure 16.

```
PROG SOFILOAD urs:30.3
HEAD FRIEDLANDER BLAST LOADING
STO#PR    27.41711758e3
STO#T0    0.268880982e-3
STO#ALF   5.83966523
STO#F1 305.015

! --- PARAMETRES DERIVES ----------------------------------------------
! Durée totale de la simulation ~ 3 périodes du 1er mode
STO#TEND  MAX(5.0/#F1,1.5*#T0)          $ T_sim = 3 / f1

! Nombre de pas pour la définition de la fonction de charge
STO#NT    1000             $ à adapter : + grand = courbe plus fine
LET#DT    #TEND/#NT        $ pas de temps pour la fonction Friedlander

LC 1 TYPE NONE TITL 'Self_Weight'
    AREA QGRP 0 TYPE PG 25*#e     $ unité kN/m²
    FUNC 0.0 1.0 ; 9999.0 1.0
! --- LC 1000 : COURBE DE FRIEDLANDER SANS PHASE NEGATIVE ------------
LC 1000 TYPE NONE TITL 'Blast Friedlander'
    AREA QGRP 0 TYPE PZZ 1.0     $ unité kN/m²
    LOOP#I #NT
       LET#t   #I*#DT          $ temps courant
       LET#tt #t/#T0         $ temps réduit t/t0

       IF #t <= #T0
          LET#P  #PR*(1.0-#tt)*EXP(-#ALF*#tt)
       ELSE
          LET#P  0.0          $ phase négative négligée
       ENDIF

       FUNC #t #P
    ENDLOOP

END
```

Figure 16. TEDDY definition of the blast pressure–time function in SOFiLOAD/DYNA (screenshot).

The blast loading is implemented through a time-dependent pressure function defined in the TEDDY input script. The script specifies the main parameters governing the blast wave, namely the peak overpressure $p_{\max}$, the positive phase duration t_+, and the functional decay of the pressure over time. The resulting pressure–time curve applied to the structure is shown in Figure 17. This pressure history is consistent with the theoretical blast representation described in Section 2.

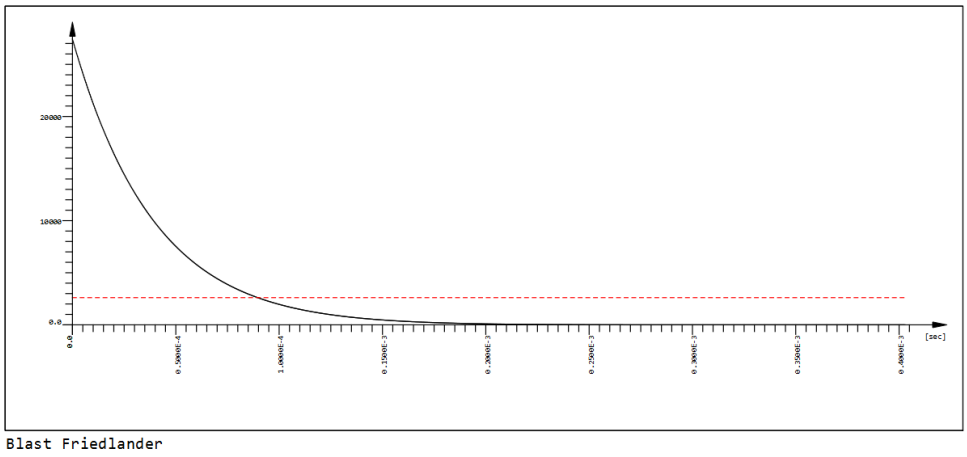


Figure 17. Resulting pressure–time history applied to the structure. The peak pressure $p_{\max}$ and positive phase duration t_+ are indicated.

For the positive phase, the blast pressure is represented using a Friedlander-type decay function of the form:

$$p(t) = p_{\max}\left(1 - \frac{t}{t_+}\right)\exp\left(-\alpha\frac{t}{t_+}\right), \qquad 0 \le t \le t_+, \tag{2}$$

where:

- $p_{\max}$ is the peak reflected overpressure,
- t_+ is the positive phase duration,
- α controls the decay rate of the exponential term.

In simplified engineering approaches, an equivalent triangular representation may also be adopted, provided that the positive impulse

$$I_+ = \int_0^{t_+} p(t)\,dt \tag{3}$$

is preserved. However, the use of a simplified triangular pulse may lead to slightly higher predicted structural responses, as the simplified representation modifies the temporal distribution of the applied pressure.

In the present study, the selected pressure representation is consistently used in both the nonlinear static equivalent approach and the nonlinear dynamic time-history analysis in order to ensure a physically meaningful comparison.

3.3 Nonlinear Dynamic Analysis Procedure

3.3.1 Equations of motion

The transient structural response is governed by the equations of motion of a multi-degree-of-freedom system:

$$\mathbf{M}\,\ddot{\mathbf{u}}(t) + \mathbf{C}\,\dot{\mathbf{u}}(t) + \mathbf{q}(\mathbf{u}(t)) = \mathbf{p}(t), \tag{4}$$

In the non-linear formulation, the linear internal force term $\mathbf{Ku}(t)$ is replaced by the non-linear internal force vector $\mathbf{q}(\mathbf{u}(t))$, which depends on the current state of deformation.

Sofistik provides both implicit and explicit time integration options for dynamic analysis [11]. In particular, non-linear explicit time integration is performed using the DYNA module, whereas implicit integration is carried out with the ASE module.

Explicit schemes require very small time steps to ensure numerical stability and are therefore typically used for short-duration phenomena such as impacts or blast loading. In contrast, implicit schemes allow for larger time steps and are generally preferred for simulations over longer time periods due to their unconditional stability.

In this study, an implicit time integration approach was adopted, even though the loading duration is short. This choice is mainly motivated by the use of the ASE module for non-linear structural analysis. This approach does not introduce significant limitations, provided that the computational cost remains acceptable.

3.3.2 Time integration and nonlinear solution

Modules ASE and DYNA describe direct time-history analysis based on step-wise time integration of the equations of motion [11].

The time axis is discretized into increments Δt, and equilibrium is enforced at discrete time steps. Within a Newmark-type framework, velocity and displacement updates are expressed as [11]:

$$\dot{\mathbf{u}}_{i+1} = \dot{\mathbf{u}}_i + \Delta t \left[(1-\gamma)\ddot{\mathbf{u}}_i + \gamma \ddot{\mathbf{u}}_{i+1} \right], \tag{5}$$

$$\mathbf{u}_{i+1} = \mathbf{u}_i + \Delta t\, \dot{\mathbf{u}}_i + \Delta t^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\mathbf{u}}_i + \beta \ddot{\mathbf{u}}_{i+1} \right]. \tag{6}$$

For non-linear dynamic analysis, it is often beneficial to introduce numerical damping in order to filter out high-frequency response components that are not accurately captured by the spatial discretization or by the chosen time step. This is achieved using the Hilber–Hughes–Taylor (α) method, which modifies the equilibrium equation while preserving the Newmark integration scheme. The equilibrium equation becomes:

$$\mathbf{M}\ddot{\mathbf{u}}_{n+1} + (1+\alpha)\mathbf{C}\dot{\mathbf{u}}_{n+1} - \alpha\mathbf{C}\dot{\mathbf{u}}_n + (1+\alpha)\mathbf{q}(\mathbf{u}_{n+1}) - \alpha\mathbf{q}(\mathbf{u}_n) = (1+\alpha)\mathbf{p}_{n+1} - \alpha\mathbf{p}_n. \tag{7}$$

The parameter α controls the amount of numerical damping, with $\alpha = 0$ corresponding to the standard Newmark method. Negative values of α introduce numerical dissipation, which is particularly useful in non-linear analyses where spurious high-frequency oscillations may occur. Recommended values typically lie in the range $-0.3 \leq \alpha \leq 0$, with associated parameters:

$$\beta = \frac{(1-\alpha)^2}{4}, \quad \gamma = \frac{1-2\alpha}{2}.$$

In this study, an implicit integration scheme is adopted using the ASE module together with a HHT-α formulation. Following the recommendations of the SOFiSTiK documentation, a numerical damping parameter corresponding to `THE` $= 0.7$ ($\alpha = 0.3$) is used. This choice provides an effective damping of non-physical high-frequency responses while maintaining overall accuracy in the non-linear dynamic analysis.

3.3.3 Static vs dynamic solution path

To enable a consistent comparison with nonlinear static analysis, the same structural model, material definitions, and load spatial distributions are used across both approaches. The main difference lies in the temporal nature of loading and the presence of inertia and damping terms in Eq. (4). The nonlinear static approach is solved with ASE [10], while the transient response is computed with DYNA [11].

3.4 Numerical Stability and Good Modelling Practice

Nonlinear dynamic simulations under blast loading are sensitive to numerical parameters and modelling assumptions. This section summarizes the key practices adopted to ensure stable and meaningful results, with emphasis on aspects highlighted in the DYNA and SOFiLOAD manuals.

3.4.1 Time step selection

The time step Δt must be small enough to resolve the pressure rise and the relevant structural response frequencies, yet large enough to keep computations tractable. DYNA notes that overly small time steps combined with consistent mass formulations may lead to oscillatory behaviour, and also highlights that, for explicit schemes, stability requires Δt to remain below a critical threshold related to wave propagation and element size [11]. In practice, Δt is selected based on:

- the shortest characteristic time of the applied blast load (rise time and positive phase),
- the structural periods of interest (dominant modes),
- mesh-dependent resolution requirements (avoid numerical artifacts).

3.4.2 Load discretization and mesh dependency

SOFiLOAD emphasizes that FE systems operate through equivalent nodal/element loads, and that the generated nodal loads can only be as accurate as the FE mesh [9]. For blast pressures applied over surfaces, this implies:

- avoiding point-load representations of spatially distributed phenomena,
- refining the mesh where load gradients or structural curvature are significant,
- maintaining consistent load mapping when comparing static and dynamic runs.

3.4.3 Damping Assumptions

For rapid dynamic phenomena such as blast loading, the relevance of damping must be carefully assessed. According to the AFGC recommendations [12], for very short-duration events (a few milliseconds), the damping term in the equation of motion is generally negligible compared to inertia and stiffness contributions.

During the positive phase of a blast, the structural response is dominated by inertial effects and nonlinear stiffness degradation. The viscous damping term $\mathbf{C}\dot{\mathbf{u}}(t)$ remains of significantly smaller magnitude than the inertial term $\mathbf{M}\ddot{\mathbf{u}}(t)$ over the short time window considered.

In this study, the analysis focuses on peak structural response immediately following the blast load. 8 Consequently, damping is neglected in the transient simulations, in accordance with AFGC recommendations, since post-shock vibration behaviour is not the primary objective.

3.4.4 Nonlinear convergence and result extraction

Nonlinear transient analyses require convergence control (iteration limits, tolerances) and careful result extraction (peak deflection, support reactions, internal forces). DYNA provides dedicated options for evaluating maxima/minima of response quantities and for storing time histories of selected degrees of freedom [11]. In this thesis, response metrics are extracted consistently across methods to support a fair comparison (time histories and peak values for displacements and key internal forces).

3.5 Model Verification

Before applying the numerical methodology to the engineering case studies, the modelling approach implemented in SOFiSTiK must be verified. The verification procedure consists in reproducing structural blast scenarios reported in the literature and comparing the numerical results obtained in this work with the published data.

3.5.1 Verification Methodology

The verification strategy is based on the reproduction of previously published experimental or numerical studies on reinforced concrete members subjected to blast loading. Two reference cases were selected from the literature: a reinforced concrete slab subjected to blast loading and a reinforced concrete beam tested under close-in explosion conditions.

The numerical models developed in SOFiSTiK reproduce the geometry, boundary conditions, material properties and loading conditions reported in the reference studies. The comparison focuses mainly on the peak displacement response and the overall structural behaviour.

A specific challenge concerns the modelling of the blast load. In the reference studies, the blast is implemented using dedicated modules available in explicit dynamic solvers such as LS-DYNA or ABAQUS, where the user only specifies the TNT equivalent mass and the stand-off distance as explaine in section 2.8. The pressure-time history is then generated internally by the software.

In contrast, SOFiSTiK requires the blast load to be defined explicitly. Therefore, the pressure-time histories used in this work are derived from the empirical blast parameters proposed by Kingery and Bulmash [1].

3.5.2 Verification case 1 : Reinforced Concrete Beam

The first verification case concerns a reinforced concrete beam subjected to blast loading. The reference study (**Zhang2013**, [13]) combines experimental observations and numerical simulations performed using ABAQUS. The objective of the present work is to reproduce this configuration using SOFiSTiK and to compare the resulting structural response.

Figure 18 illustrates the experimental setup used in the reference study. The reinforced concrete beam is subjected to a blast load generated by a TNT charge located at a given stand-off distance. The supports and boundary conditions reproduce a simply supported beam configuration, allowing the structural response to be dominated by flexural behaviour.

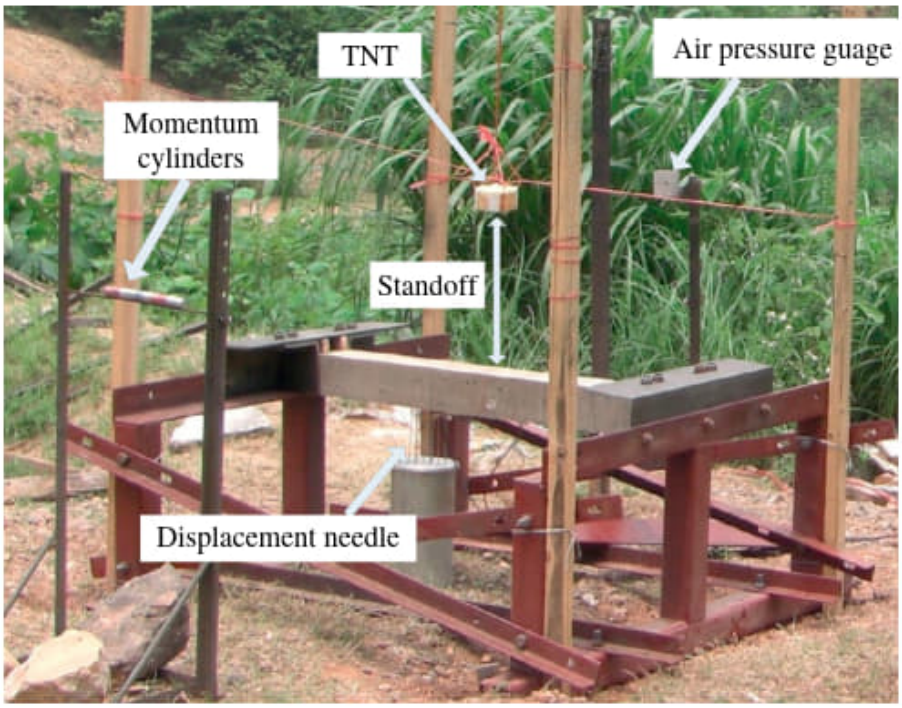


Figure 18. Experimental setup of the reinforced concrete beam subjected to blast loading.(adapted from **Zhang20135**).

The geometry, boundary conditions and material properties reported in the reference study were reproduced in the SOFiSTiK finite element model. However, a difference exists between the modelling strategies adopted in the two numerical approaches. In the reference numerical study, the beam is modelled using three-dimensional solid elements in ABAQUS, allowing a detailed representation of the concrete volume and reinforcement layout. In contrast, the present work uses a shell-based modelling approach in SOFiSTiK, where the beam is represented using QUAD finite elements. This approach significantly reduces the computational cost while still capturing the global structural response.

The reinforcement distribution implemented in the SOFiSTiK model is defined in terms of equivalent steel areas (in cm^2/m) assigned to the QUAD elements forming the beam. These values represent the smeared reinforcement in each direction and layer, as described in Section 3.1.2. Figure 19 illustrates the resulting spatial distribution of reinforcement within the finite element mesh. The reinforcement values assigned to the QUAD elements

are derived from the reinforcement configuration reported in the experimental study. The tensile, compressive and hoop reinforcement bars are specified as HPB 235 steel with a diameter of 6 mm.

Assuming two reinforcement bars in both the top and bottom layers of the beam, the equivalent reinforcement area per unit width can be calculated from the cross-sectional area of the bars. The area of a single bar is given by:

$$A_{\text{bar}} = \pi \left(\frac{d}{2}\right)^2 = \pi(0.3cm)^2$$

Considering two bars distributed over a beam width of 0.1 m, the equivalent smeared reinforcement is:

$$A_s = \frac{2\,A_{\text{bar}}}{0.1} = 5.65 \quad cm^2/m$$

This leads to a reinforcement value consistent with the one implemented in the SOFiSTiK model using the BEMESS module. The resulting reinforcement distribution, expressed in cm^2/m, is shown in Figure 19.

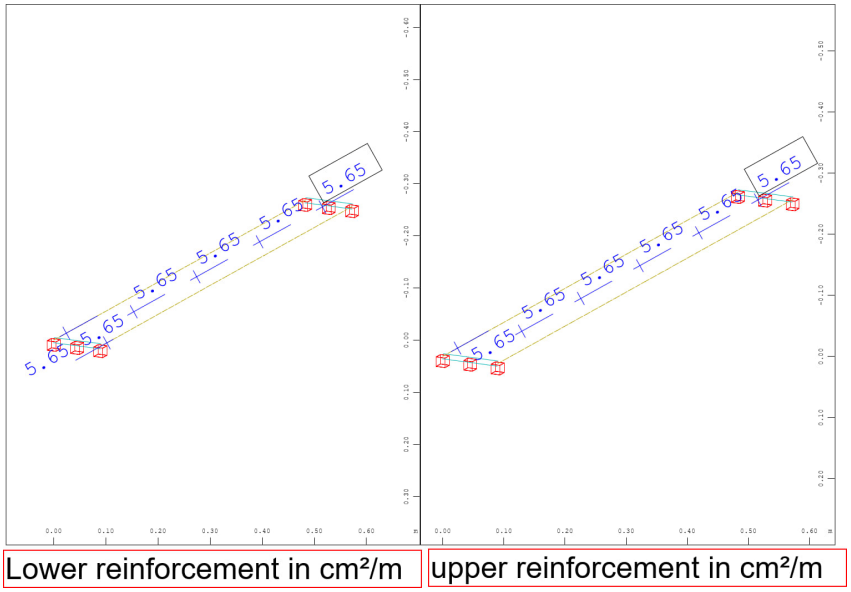
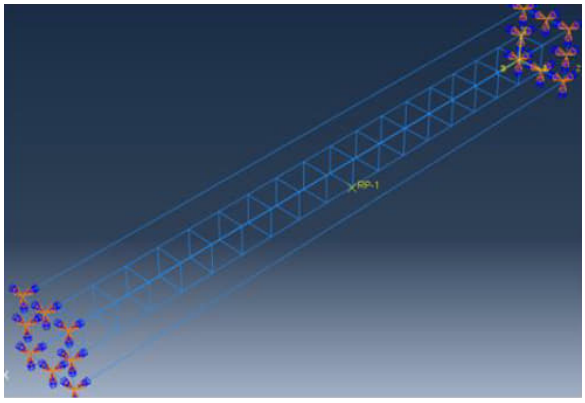
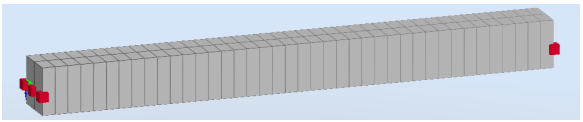


Figure 19. Reinforcement layout implemented in the SOFiSTiK beam model (value in cm^2/m in the QUAD element.



Reference model (ABAQUS) (adapted from [13])



SOFiSTiK finite element model

Figure 20. Comparison between the beam model from the reference study and the SOFiSTiK model.

The blast loading was reconstructed using the Kingery–Bulmash blast correlations and applied to the beam surface using the SOFiSTiK loading module.

Figure 21 presents the deformation obtained in the SOFiSTiK simulation.

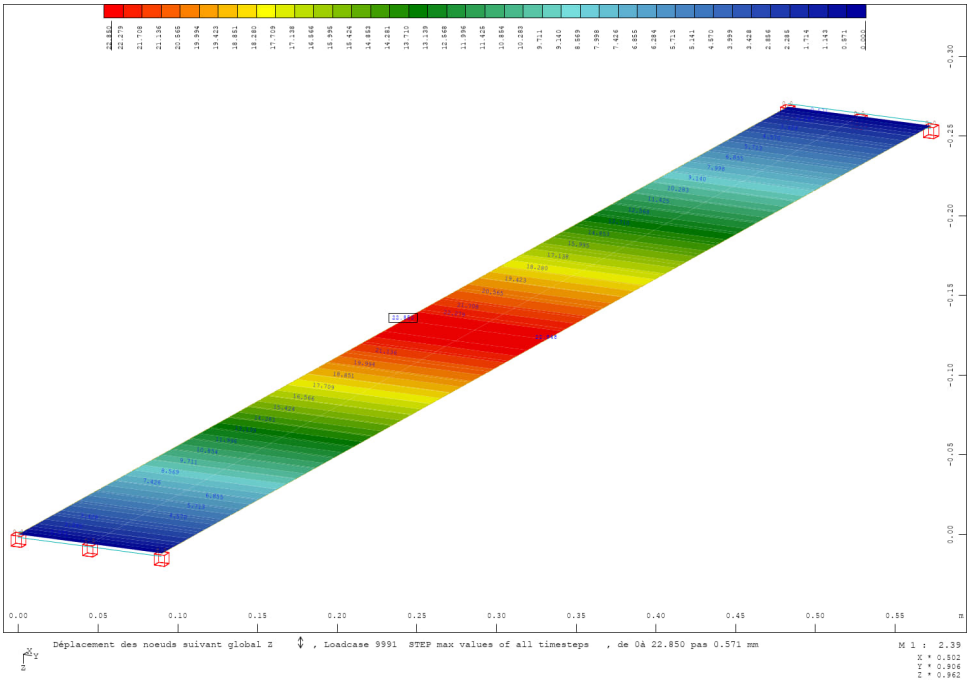


Figure 21. Maximum vertical displacement obtained in the SOFiSTiK simulation.

The structural response obtained in the SOFiSTiK simulation shows a typical flexural behaviour, with the maximum displacement occurring at mid-span of the beam.

Table 1 compares the maximum mid-span displacement obtained experimentally, in the ABAQUS numerical model, and in the present SOFiSTiK simulation.

Table 1. Comparison of maximum mid-span displacement

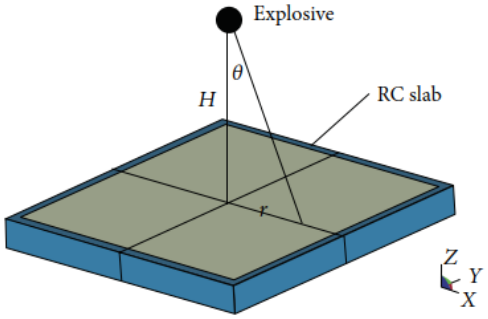
Method	Maximum displacement (cm)	Relative difference (%)
Experiment	25	–
ABAQUS simulation	26	+4.0
SOFiSTiK simulation	23	-8.0

The results obtained using SOFiSTiK remain in good agreement with both the experimental observations and the ABAQUS simulation. Although the SOFiSTiK model predicts a slightly smaller displacement, the relative difference remains limited. This discrepancy may be partly attributed to differences in the modelling assumptions, in particular the use of shell QUAD elements instead of fully three-dimensional solid elements. Additional sources of deviation may also arise from uncertainties in the definition of boundary conditions and material properties, which may not exactly match those of the experimental setup or the reference numerical model. These factors can influence the predicted structural response and should therefore be considered when interpreting the differences between the simulations and the experimental results.

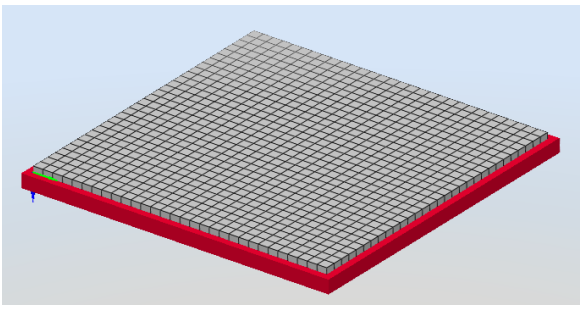
3.5.3 Verification case 2 : Reinforced Concrete Slab

The second verification case corresponds to the reinforced concrete slab studied by Talaat et al. [3]. The study investigates the dynamic response of a simply supported RC slab subjected to blast loading generated by a TNT detonation above the slab.

The geometry, materials and support conditions described in the reference study were reproduced in the SOFiSTiK finite element model.



Reference configuration (adapted from Talaat et al. [3]



SOFiSTiK FE model

Figure 22. Comparison between the reference slab configuration and the SOFiSTiK model.

The reinforcement layout implemented in the SOFiSTiK model follows the configuration reported in the reference study. Figure 23 illustrates the reinforcement arrangement for the upper and lower layers in both orthogonal directions. The reinforcement is distributed in two orthogonal meshes in the top and bottom layers of the slab, in agreement with the reinforcement scheme described in the reference article.

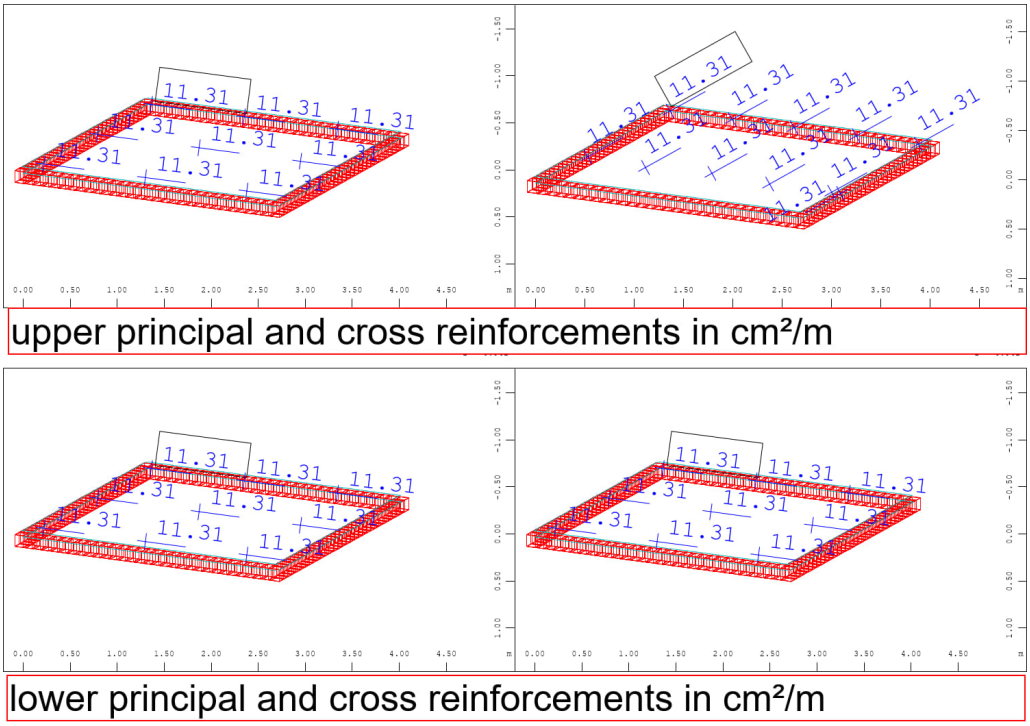


Figure 23. Reinforcement layout of the slab in the SOFiSTiK model: top and bottom reinforcement layers in both directions.

The blast loading was reconstructed using the Kingery–Bulmash relationships, which allowed the pressure–time history to be derived from the TNT equivalent mass and stand-off distance reported in the reference study.

Figure 24 shows the vertical displacement field obtained at the time corresponding to the peak response of the slab.

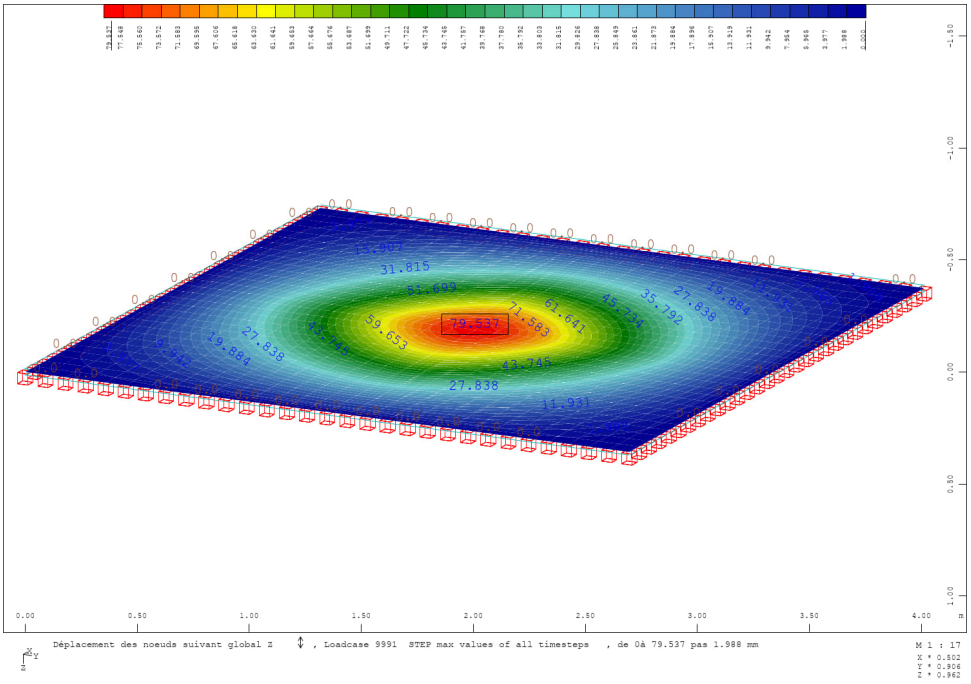


Figure 24. Vertical displacement u_z of the slab at peak response in the SOFiSTiK simulation.

Figure 25 compares the mid-span displacement histories obtained with SOFiSTiK and

the LS-DYNA results reported in the literature.

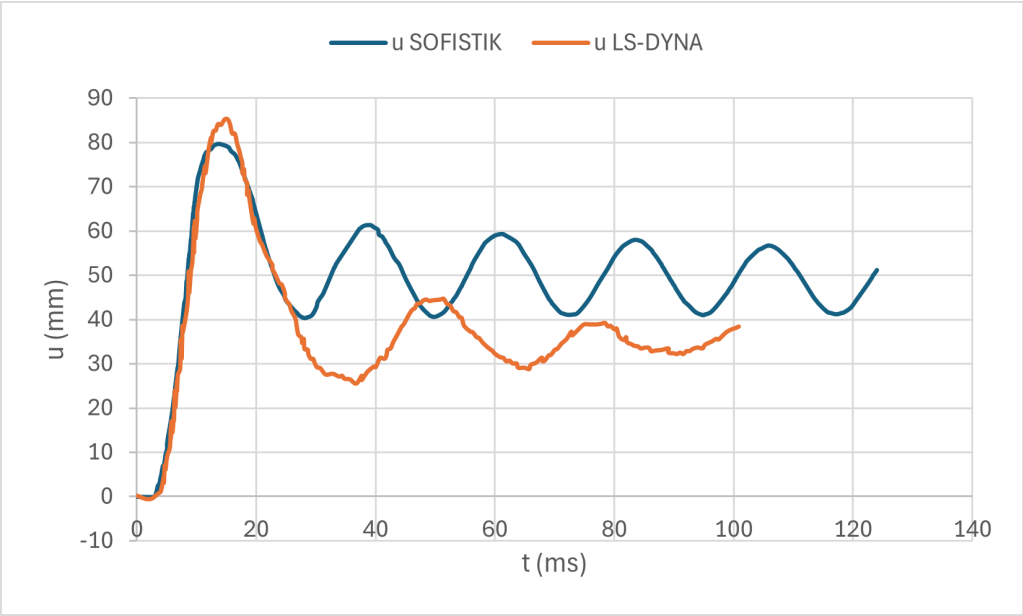


Figure 25. Comparison of mid-span displacement histories obtained with SOFiSTiK and LS-DYNA.

The comparison shows a good agreement for both the peak displacement and the initial dynamic response. The first part of the response, which corresponds to the direct structural reaction to the blast impulse, is very similar in the two simulations.

Some differences appear in the subsequent oscillations of the displacement history. These differences may be attributed to several factors, including differences in numerical formulation, damping assumptions, and blast load implementation between the two software packages. In addition, the present model represents the slab using shell-type QUAD elements, whereas the reference numerical model relies on a three-dimensional solid representation of the concrete volume. This difference in modelling strategy may also contribute to the observed discrepancies in the later stages of the dynamic response.

Despite these differences, the overall agreement between the two displacement histories confirms that the SOFiSTiK model is able to reproduce the main features of the dynamic structural response under blast loading. This comparison highlights the potential of dynamic finite element analyses as a reliable tool for real design applications, where simplified equivalent static loads are often used for structural dimensioning.

4.1 Case Study 1: Blast Assessment of a Digester Structure

4.1.1 Project Background

The first case study concerns the blast assessment of anaerobic digesters designed for the Seine Aval wastewater treatment plant operated by the SIAAP (Service Public de l'Assainissement Francilien). This facility, located in the Yvelines region near Paris, treats wastewater from approximately six million inhabitants and is one of the largest wastewater treatment plants in Europe.

As part of the modernization program of the biogas treatment installations, several studies were carried out by INERIS to evaluate potential explosion scenarios within the digesters. These studies investigated different roof configurations, including vented roofs and dome structures composed of opening petals designed to relieve internal overpressure during an explosion.

Subsequent investigations aimed at refining the modelling assumptions by considering more detailed geometric data and improved representations of the opening mechanisms of the roof petals.

In the present work, a simplified finite element model of the digester structure was developed in SOFiSTiK in order to assess its structural response under blast loading conditions.

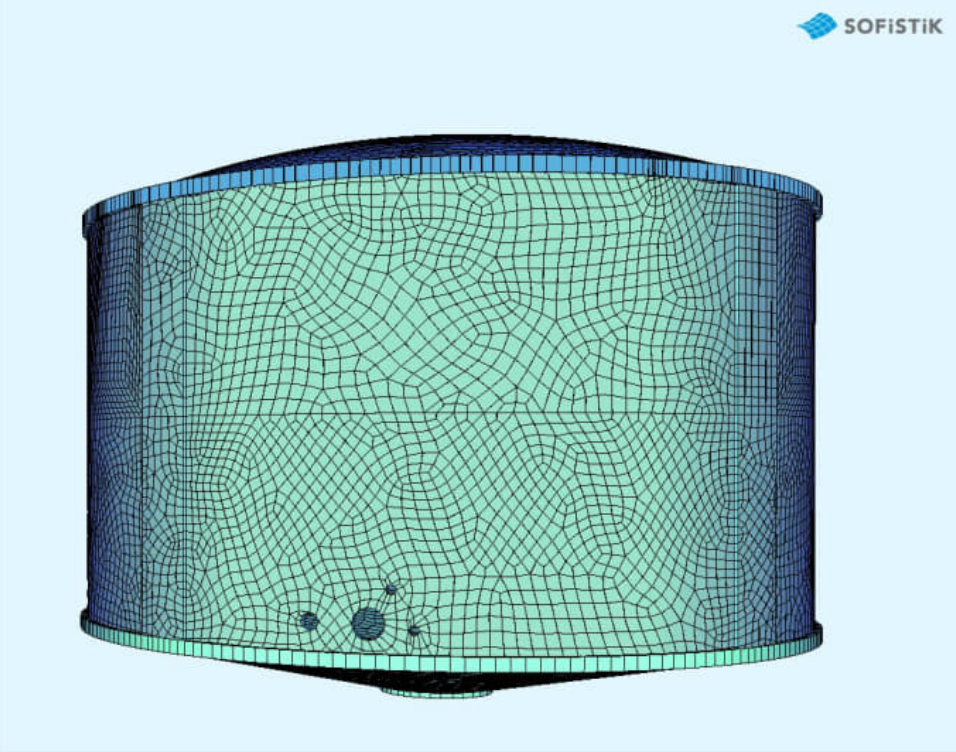


Figure 26. Finite element model of the digester structure used for the blast analysis.

4.1.2 Blast Loading Scenario

The explosion scenario considered in this case study corresponds to an internal deflagration of biogas within the digester. The pressure evolution associated with this event was evaluated by INERIS (French National Institute for Industrial Environment and Risks) using the EFFEX simulation tool, which estimates the combustion of the gas mixture and the resulting overpressure inside the structure.

The explosion leads to the ejection of a mixture of burnt and unburnt gases through the roof openings. The internal pressure development obtained from the EFFEX calculations is used as input for the structural analysis.

Figure 27 presents the pressure evolution inside the digester as well as the corresponding mass of gases ejected during the event. This pressure–time history is directly applied to the internal surfaces of the numerical model.

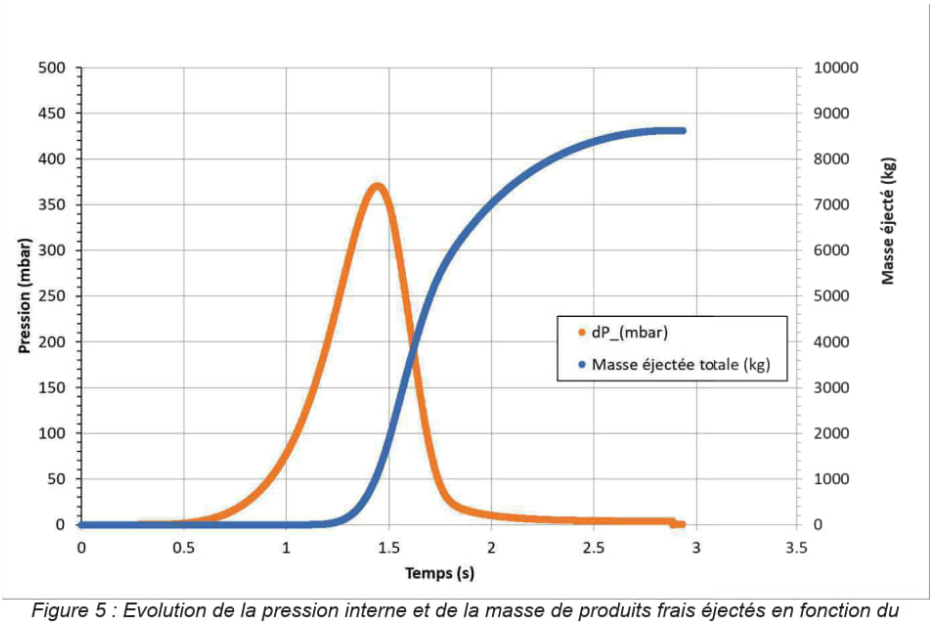


Figure 27. Pressure evolution inside the digester during the internal gas deflagration (INERIS simulation).

4.1.3 Finite Element Model Description

The digester structure is entirely made of reinforced concrete and is modelled accordingly using QUAD finite elements. These elements allow for a coupled representation of membrane and bending behaviour, which is essential for capturing the global response of the cylindrical wall and dome under internal pressure loading.

The reinforcement is introduced within the QUAD elements as smeared layers defined by equivalent steel areas (in cm^2/m) in each direction, as described in Section 3.1.2. This approach enables the representation of the composite behaviour of reinforced concrete while maintaining computational efficiency for large-scale structural analyses.

The structure is assumed to be simply supported on the ground. The soil–structure interaction is modelled using distributed vertical springs applied under the entire lower dome of the digester. The stiffness of these springs is derived from the geotechnical

properties obtained from the soil investigation carried out during the initial design phase of the project. This modelling approach allows the flexibility of the foundation system to be taken into account without explicitly modelling the surrounding soil.

In the initial design study, the structural response was assessed using a nonlinear static analysis. A uniform internal pressure corresponding to the maximum overpressure of the explosion (350 mbar see section 4.1.4) was applied to the entire internal surface of the digester. This simplified approach neglects the transient nature of the loading and the interaction between pressure evolution and structural deformation.

In the present work, this initial static approach is compared with a nonlinear dynamic analysis in which the time-dependent pressure evolution is explicitly considered.

4.1.4 Non linear Static Analysis

In the first approach, the explosion load is represented using an equivalent static pressure applied uniformly on the internal surfaces of the digester. The value of this pressure corresponds to the maximum overpressure predicted by the INERIS deflagration analysis.

This simplified method neglects the temporal evolution of the pressure and assumes that the peak pressure acts simultaneously on the entire structure. The load is therefore applied as a constant pressure on the internal surfaces of the dome and cylindrical wall.

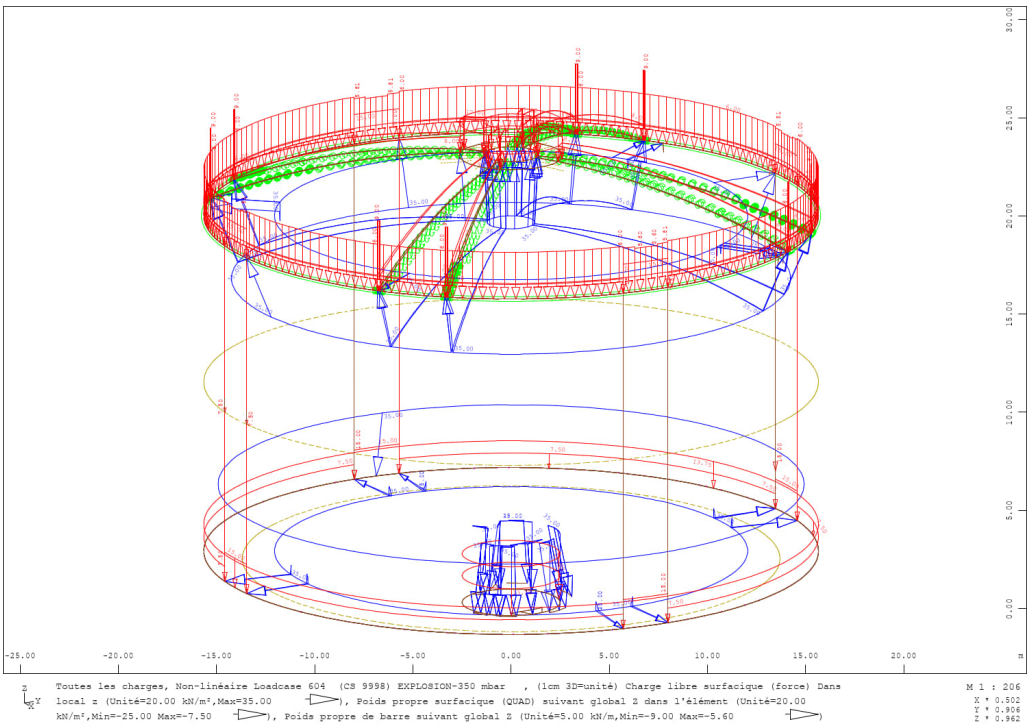


Figure 28. Equivalent static pressure applied on the internal surfaces of the digester in blue (and selfweight in red).

Under this assumption, the structural response corresponds to the most conservative loading condition since the pressure relief caused by the opening of the roof petals is not considered.

4.1.5 Nonlinear Dynamic Analysis

In the nonlinear dynamic analysis, the explosion load is introduced as a pressure–time history derived from the INERIS deflagration calculations. The temporal evolution of the internal pressure is therefore explicitly considered during the simulation.

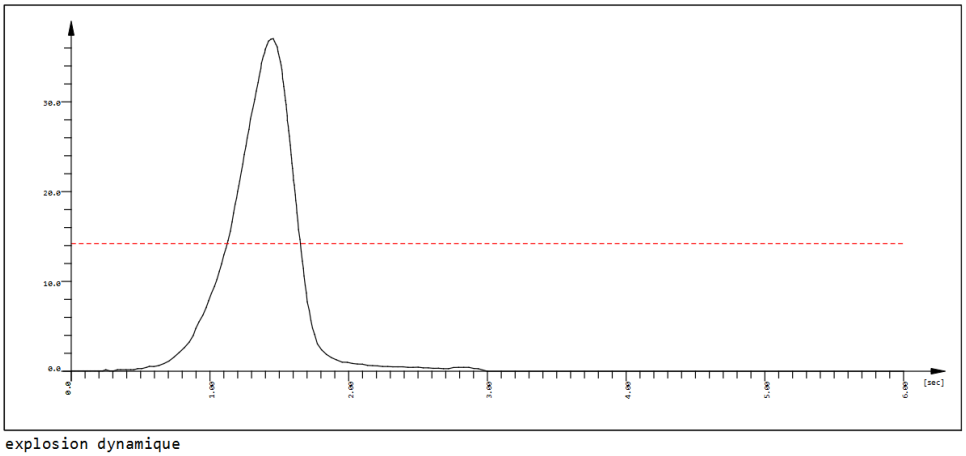


Figure 29. Pressure–time history applied in the nonlinear dynamic analysis.

The progressive opening of the roof petals during the explosion allows part of the combustion gases to escape, which reduces the internal pressure acting on the structure. This phenomenon is captured in the dynamic simulation and directly influences the structural response.

4.1.6 Comparison of Results

Figures 30 and 31 present the deformation of the digester roof obtained respectively from the equivalent static and nonlinear dynamic analyses.

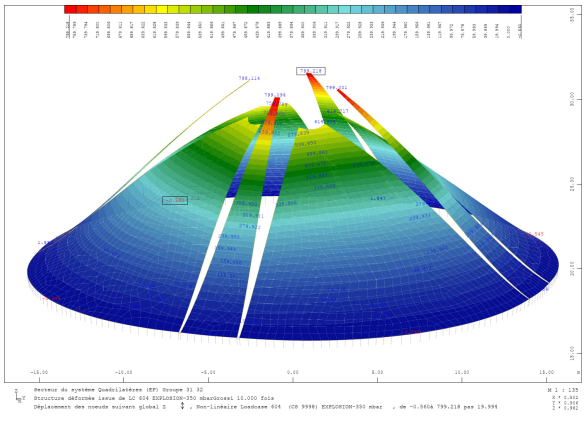


Figure 30. Petal opening obtained with the equivalent static analysis.

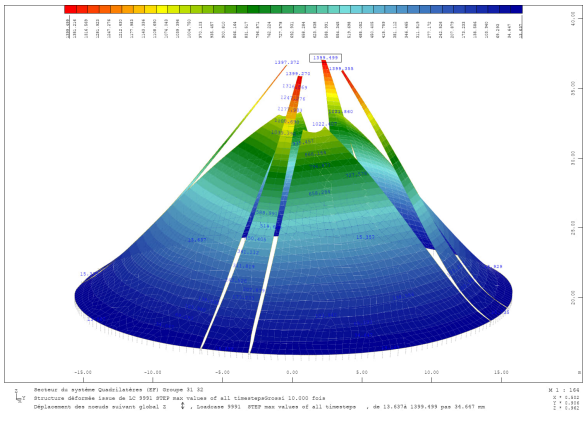


Figure 31. Petal opening obtained with the nonlinear dynamic analysis.

The results highlight the influence of the loading approach on the structural response. In the equivalent static analysis, the maximum pressure is applied simultaneously over the entire structure.

In contrast, the nonlinear dynamic analysis accounts for the temporal evolution of the pressure and the progressive release of gases through the roof openings. This allows the interaction between the structural deformation and the pressure evolution to be captured more realistically.

In order to further analyse the structural behaviour, Figure 32 presents the maximum displacement at the tip of the roof petal as a function of the applied static overpressure.

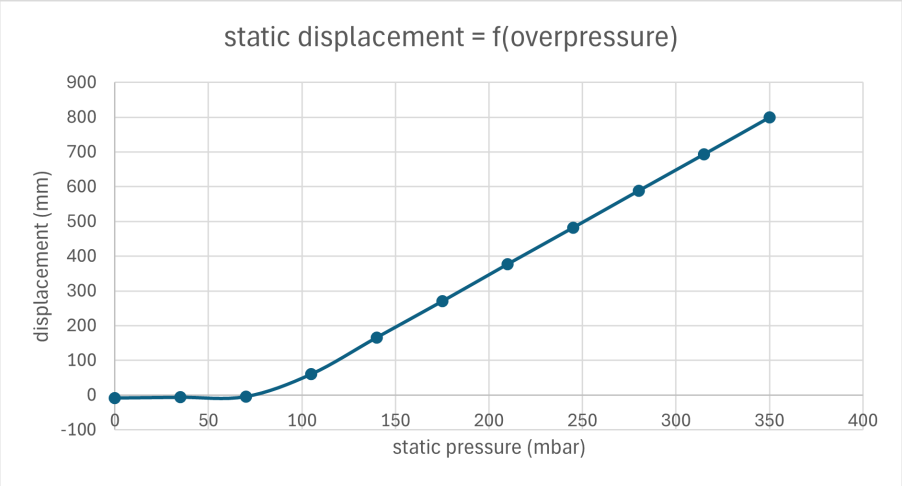


Figure 32. Maximum displacement at the tip of the roof petal as a function of the applied static overpressure.

The curve shows that, beyond the initial activation of the opening mechanism, the displacement evolves approximately linearly with the applied pressure. This indicates that the global structural behaviour remains predominantly linear once the petals start opening, with no significant material or geometric instability. This static response provides a reference state for interpreting the dynamic results.

Figure 33 presents the time evolution of the displacement at the tip of the roof petal obtained from the nonlinear dynamic analysis.

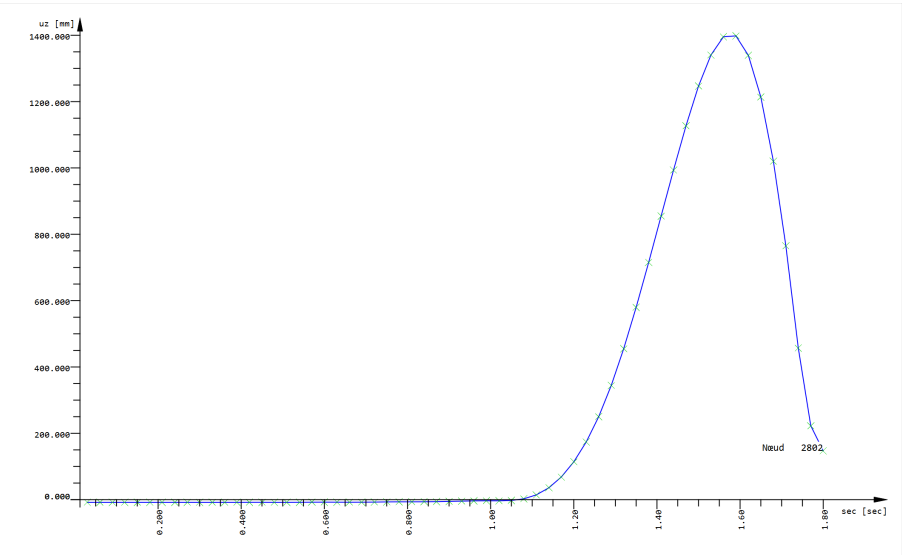


Figure 33. Time evolution of the displacement at the tip of the roof petal in the nonlinear dynamic analysis.

The dynamic response shows a progressive increase of displacement followed by a peak and a subsequent decrease, reflecting the transient nature of the loading. The maximum displacement is reached when the internal pressure is at its highest and before significant venting reduces the loading.

Table 2 summarizes the maximum displacement obtained at the tip of the roof petal for both approaches.

Table 2. Maximum displacement at the tip of the digester roof petal.

Analysis method	Maximum displacement (mm)
Equivalent static analysis	799
Dynamic analysis	1399

The results show that the dynamic analysis predicts a significantly larger opening of the petals. This difference illustrates the importance of considering the transient interaction between structural deformation and internal pressure evolution during the explosion.

Although the static response exhibits an approximately linear behaviour with respect to the applied pressure, it does not capture the coupling between pressure evolution and structural deformation. The dynamic analysis therefore provides a more realistic representation of the physical phenomenon, particularly the progressive opening of the roof and the associated pressure relief.

4.2 Case Study 2: Slab subjected drop of a heavy container

4.2.1 Project Background

The study focuses on assessing the structural response of a reinforced-concrete protection slab subjected to the accidental drop of a heavy container. This scenario corresponds to a 1 m fall of a standard container during handling operations above a pedestrian pathway.

Although this loading case does not correspond to a blast event, it still involves a very rapid and highly dynamic loading condition. Such impact scenarios share several characteristics with blast loading, including short-duration load application and significant inertial effects in the structural response. For this reason, similar dynamic analysis approaches can be used to evaluate the structural behaviour.

A simplified dynamic approach is required to better capture impact mechanisms, including the influence of impact location and the distribution of energy between the container and the structure. The analysis includes dynamic modelling of the slab and its supporting system (Fig. 35), evaluation of structural stresses and damage criteria, and optimisation of the reinforcement design where necessary.

The objective is to verify the structural robustness of the protective slab under accidental impact and ensure adequate safety for public use.

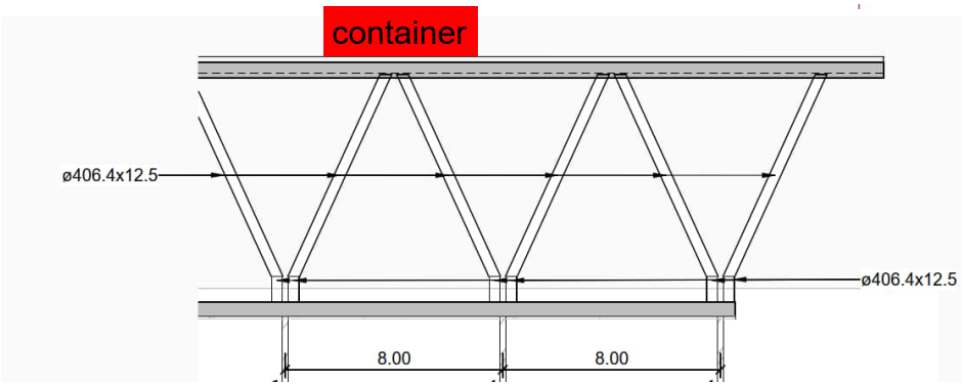


Figure 34. Schema drop of the container

4.2.2 Finite Element Model Description

The protective slab is entirely made of reinforced concrete and is modelled using QUAD finite elements. These elements allow for an accurate representation of both bending and membrane behaviour, which is essential for capturing the response of the slab under impact loading. The reinforcement is introduced as equivalent distributed steel areas within the elements, consistent with the modelling approach described previously.

In reality, the slab extends over a significantly larger area. However, in order to limit the computational cost, only a representative portion of the structure is modelled. The selected dimensions are sufficient to avoid boundary effects influencing the local response under impact. This modelling choice is slightly conservative, as it restricts the redistribution of forces over larger distances.

The slab is supported by tubular steel columns with a diameter of 406 mm and a thickness of 12.5 mm. These columns are modelled with a linear elastic behaviour, as their response remains within the elastic range for the considered loading scenario. In contrast, the slab is modelled using nonlinear material behaviour in order to capture cracking and energy dissipation mechanisms.

For the static analysis, the impact is represented by an equivalent static load applied over the contact area of the container. In the dynamic analysis, the impact is modelled through a contact interaction between the container and the slab (see section 4.2.3). The contact is defined on the container surface, which is assumed to be the stiffest component of the system. Alternative modelling using a distributed contact over the full contact area was also investigated and led to similar results.

A simplifying assumption is made that the container impacts the slab with a flat orientation. This assumption is considered conservative for design purposes, as it maximises the contact area and the resulting load transfer to the structure.

4.2.3 Loading Scenario

The second case study investigates the accidental impact of a 30 t container on the structural slab. The impact velocity is estimated by assuming a free fall from a height of 1 m. Neglecting air resistance, the impact velocity is obtained from:

$$v = \sqrt{2gh} \tag{8}$$

where $g = 9.81\,\mathrm{m/s^2}$ and $h = 1\,\mathrm{m}$. This leads to an impact velocity of approximately $4.5\,\mathrm{m/s}$.

The structural model adopted for the simulation is shown in Figure 35. The container is represented as a concentrated impacting mass acting on the slab.

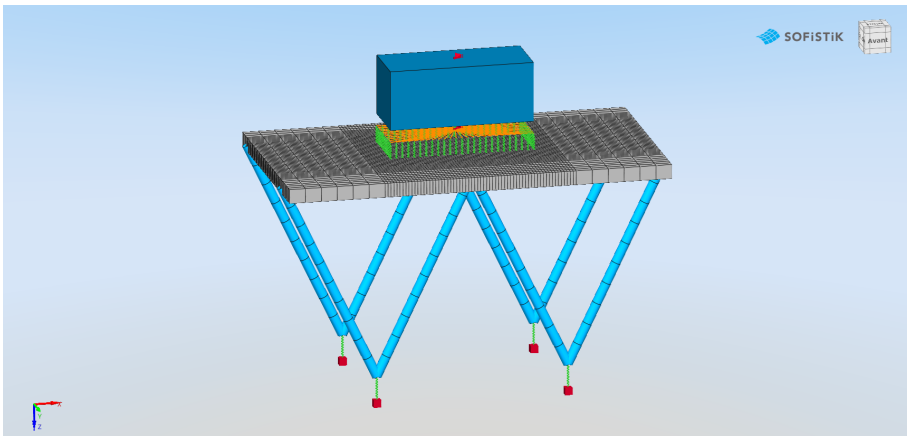


Figure 35. Finite element model used for the container impact scenario.

According to Eurocode 1 [6], a simplified estimate of the maximum impact force can be obtained from:

$$F = v\sqrt{km} \tag{9}$$

where m is the impacting mass and k is the equivalent stiffness of the impacted system. In the present study, the stiffness was estimated using the simplified expression $k = EA/L$, leading to a value of $1.0 \times 10^{10}\,\mathrm{N/m}$.

The force–time history applied in the numerical model is shown in Figure 36. From this curve, the peak impact force is approximately $50\,\mathrm{MN}$ and the impact duration is about $5\,\mathrm{ms}$.

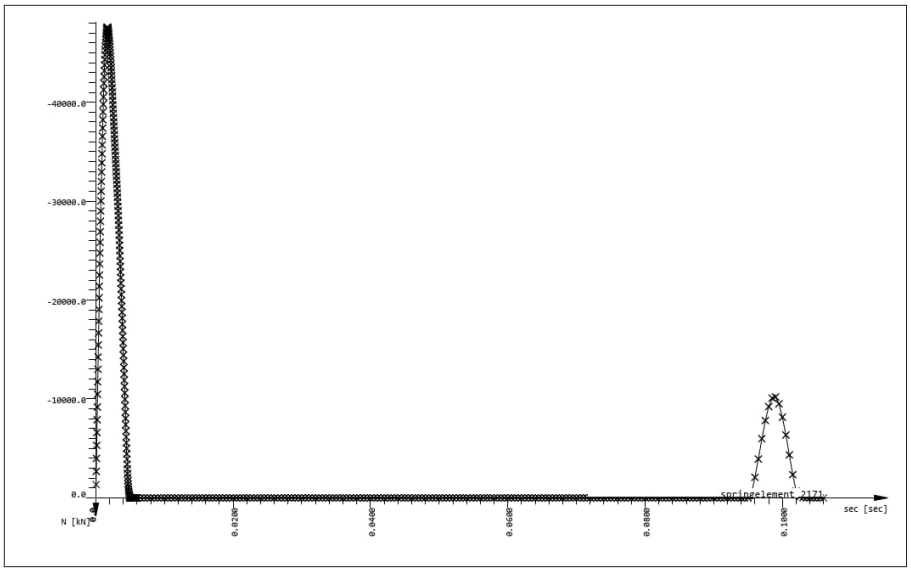


Figure 36. Impact force time history used in the dynamic simulation.

Table 3 compares the simplified Eurocode estimate with the values retained in the numerical model.

Table 3. Comparison between input parameters and Eurocode-based estimates used in the numerical model.

	Type	Eurocode / analytical value	Numerical model value
Impact mass m	Input parameter	30 000 kg	30 000 kg
Impact velocity v_0	Input parameter	4.5 m/s	4.5 m/s
Equivalent stiffness k	Input parameter	1.0×10^{10} N/m	1.0×10^{10} N/m
Maximum force $F_{\max}$	Calculated value	77.9 MN	50 MN
Impact duration Δt	Calculated value	2 ms	≈ 5 ms

In Table 3, the impact mass m, velocity v_0, and equivalent stiffness k are defined as input parameters of the problem, either directly prescribed or estimated from structural properties. In contrast, the maximum impact force $F_{\max}$ and the impact duration Δt are derived quantities, obtained from the simplified relationships proposed in Eurocode 1 or directly extracted from the dynamic simulation results.

The comparison shows that the numerical loading remains consistent with the simplified Eurocode estimate, with the same order of magnitude for the peak force. The lower peak force obtained in the model is explained by the finite duration of the impact and the simplified representation of the contact interaction.

4.2.4 Comparison of Results

The structural response obtained from the different modelling approaches is compared in this section. Four types of analyses are considered: linear dynamic analysis, nonlinear dynamic analysis, linear static analysis based on an equivalent peak load applied over the container contact surface, and nonlinear static analysis. The nonlinear static analysis did not converge, highlighting the limitations of representing a highly dynamic impact event through an equivalent static load. Such an approach tends to produce excessively con-

servative structural demands that do not reflect the actual transient interaction between the impacting mass and the structure.

Figures 37 and 42 present the displacement and bending moment distributions obtained with the linear dynamic model.

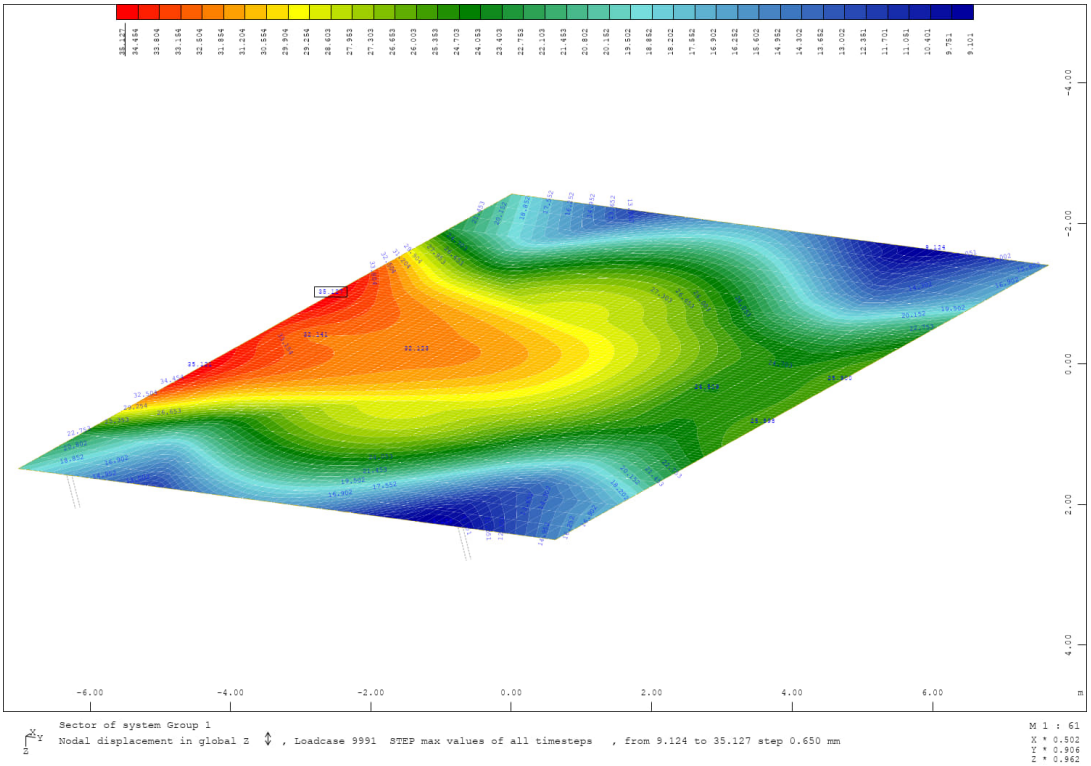


Figure 37. Maximum displacement obtained from the linear dynamic analysis.

The results obtained with the nonlinear dynamic analysis are shown in Figures 38 and 43.

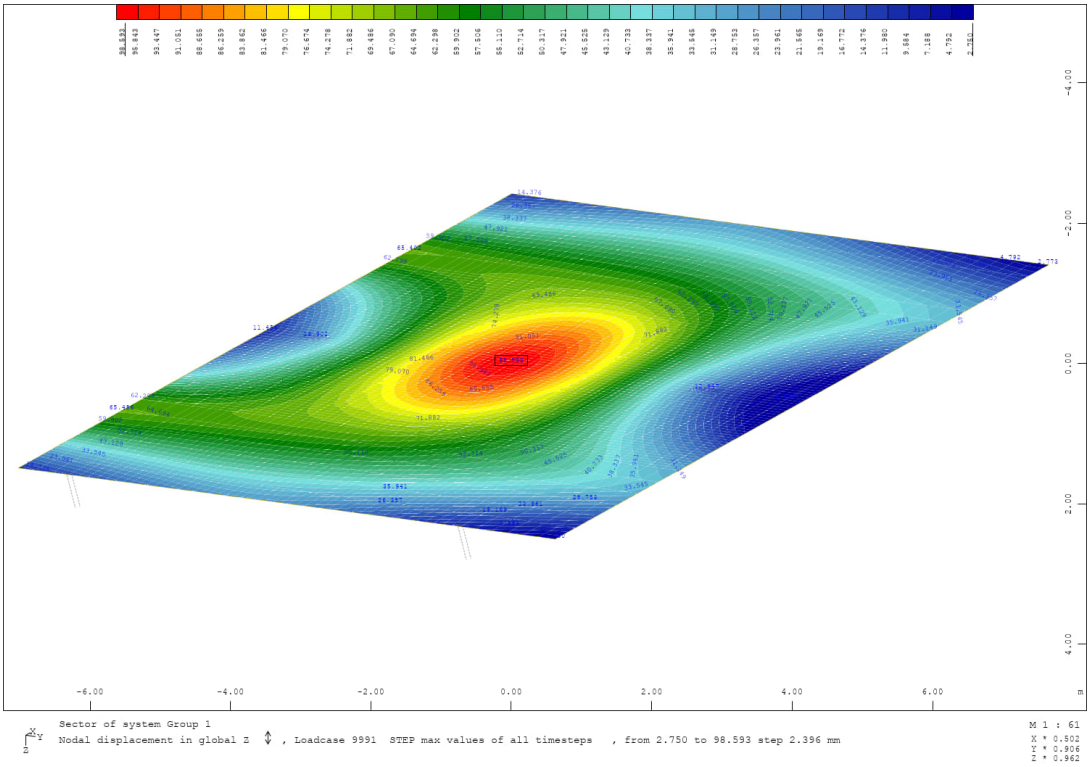


Figure 38. Maximum displacement obtained from the nonlinear dynamic analysis.

For comparison purposes, an equivalent static load corresponding to the peak dynamic force was also applied on the container contact surface. Figure 39 illustrates the applied static loading configuration.

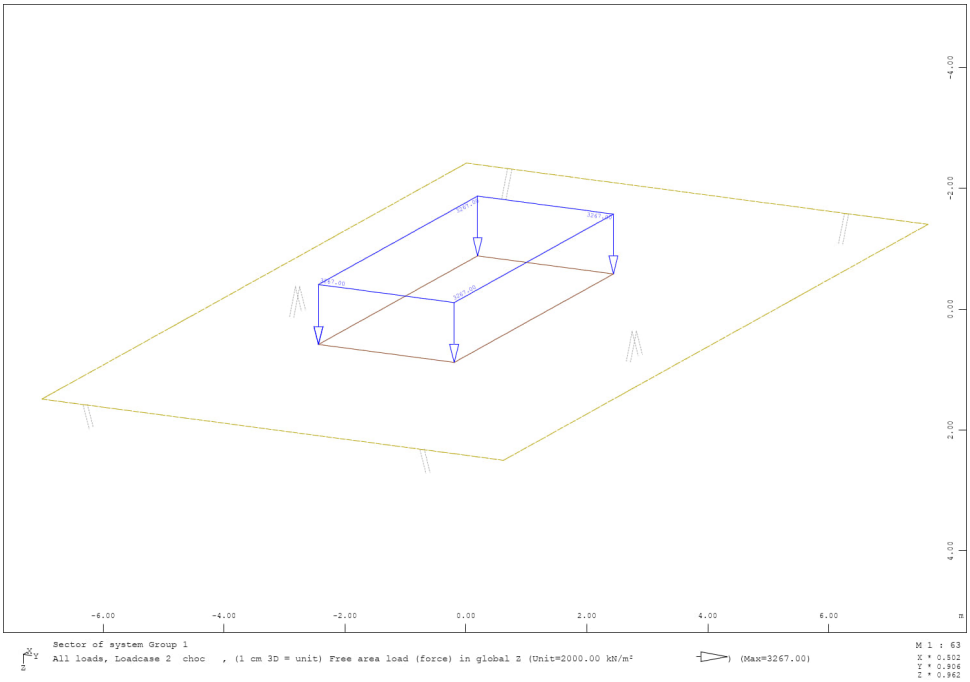


Figure 39. Equivalent static load applied on the container contact surface.

The resulting vertical displacement and bending moment distributions obtained from the static analysis are presented in Figures 40 and 44.

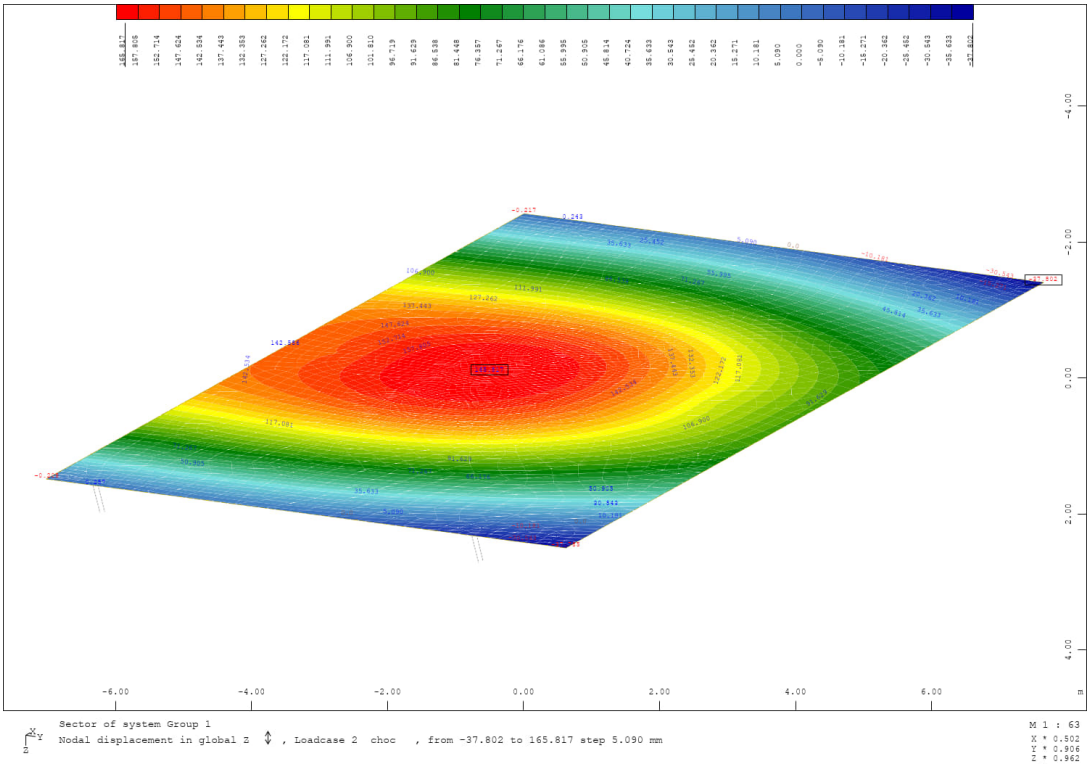
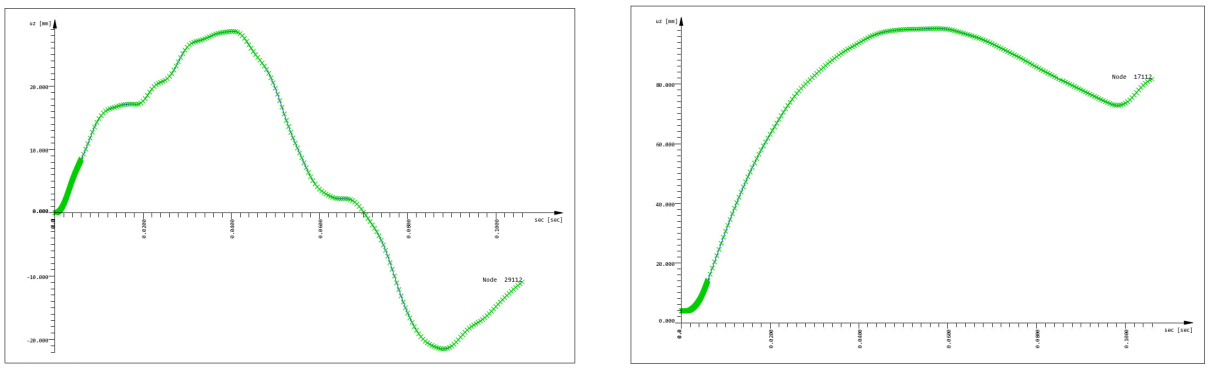


Figure 40. Vertical displacement obtained from the static analysis.

It is worth noting that the location of the maximum displacement differs between the analyses. Due to the asymmetry of the support conditions, with stiffer piles on one side, the linear dynamic analysis shows a maximum displacement near the slab edge on the less stiff side.

In contrast, the nonlinear analyses predict a maximum displacement closer to the centre of the slab, as nonlinear effects allow for a more global deformation and load redistribution.

The vertical displacement at the centre of the slab as a function of time is presented for both the linear and nonlinear dynamic analyses in Figures 41a and 41b.



(a) Linear dynamic analysis

(b) Nonlinear dynamic analysis

Figure 41. Time history of vertical displacement at the centre of the slab.

The displacement time histories at the centre of the slab highlight two distinct behaviours.

In the linear dynamic analysis, the response initially increases rapidly and reaches a maximum displacement at the slab centre. From approximately $t = 0.04\,\mathrm{s}$ onward, the

displacement at the centre starts to decrease, which indicates a redistribution of the structural response toward the slab edge on the side where the piles are less stiff. This evolution is consistent with the asymmetric stiffness of the support system: once the impact-induced deformation develops, the less restrained side attracts a larger share of the global deformation. As a result, the central displacement decreases and eventually becomes upward, while the maximum displacement is transferred away from the centre.

In contrast, the nonlinear dynamic analysis shows a more regular and physically consistent evolution of the central displacement. The response increases rapidly at the beginning of the impact, then continues to grow more slowly until reaching its maximum value. This progressive evolution reflects the combined effects of stiffness degradation and energy dissipation in the nonlinear model, leading to a smoother global deformation of the slab without the same redistribution of displacement observed in the linear analysis.

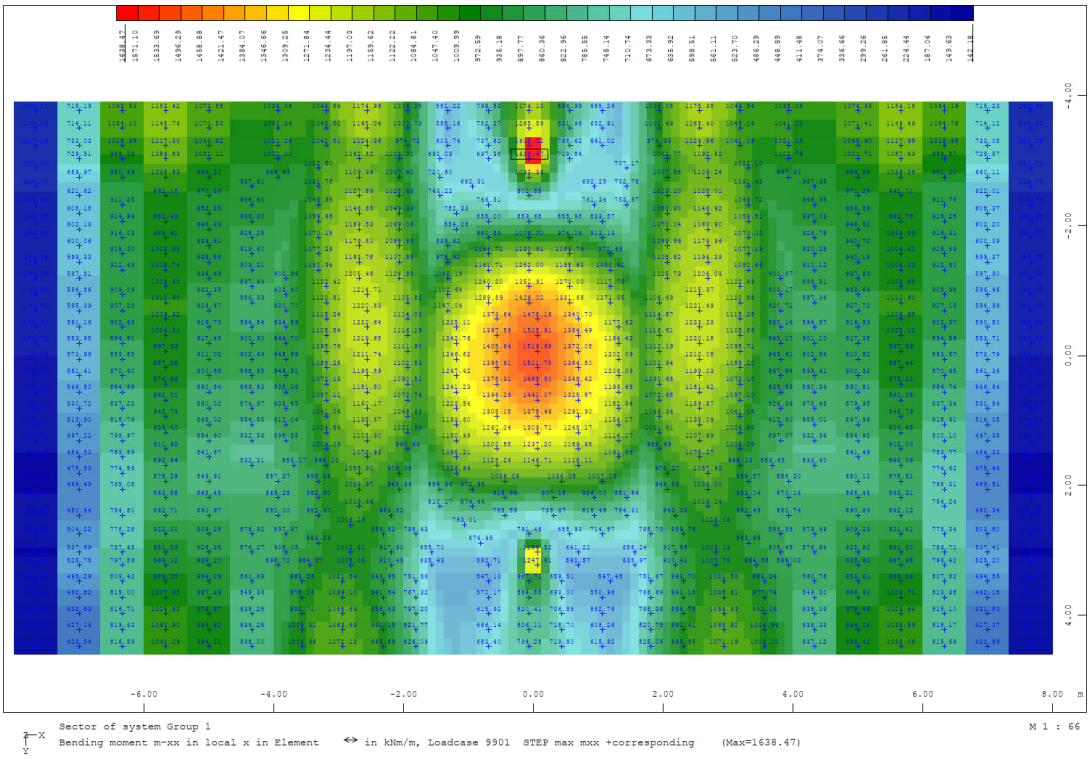


Figure 42. Bending moment distribution obtained from the linear dynamic analysis.

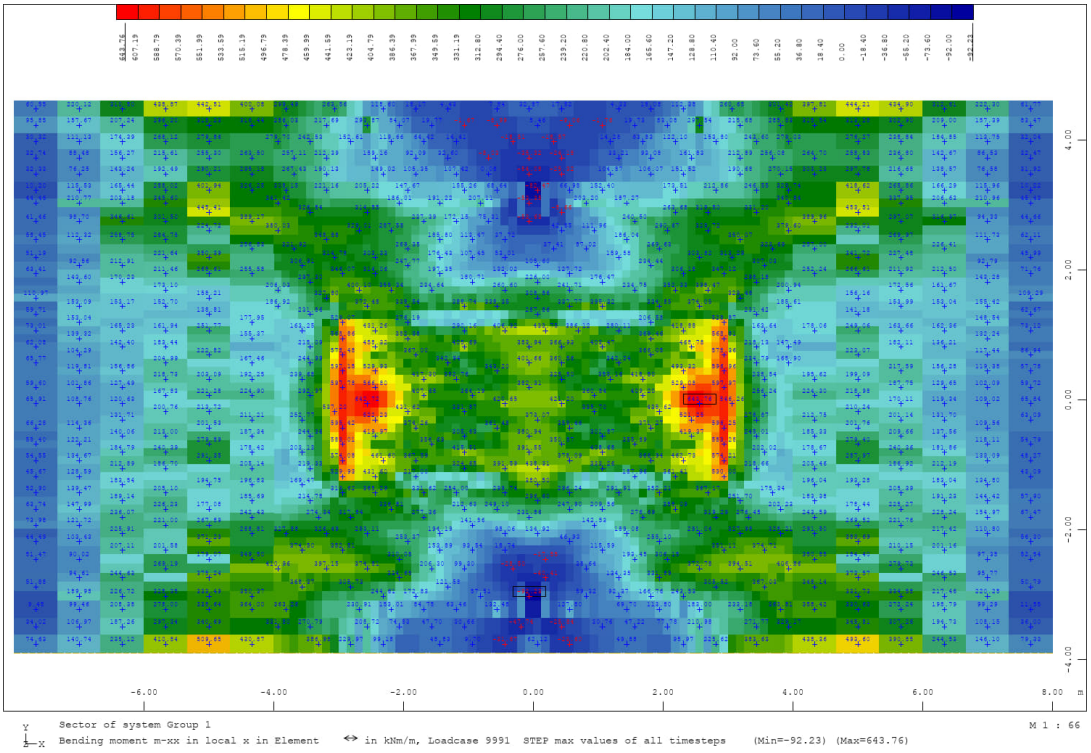


Figure 43. Bending moment distribution obtained from the nonlinear dynamic analysis.

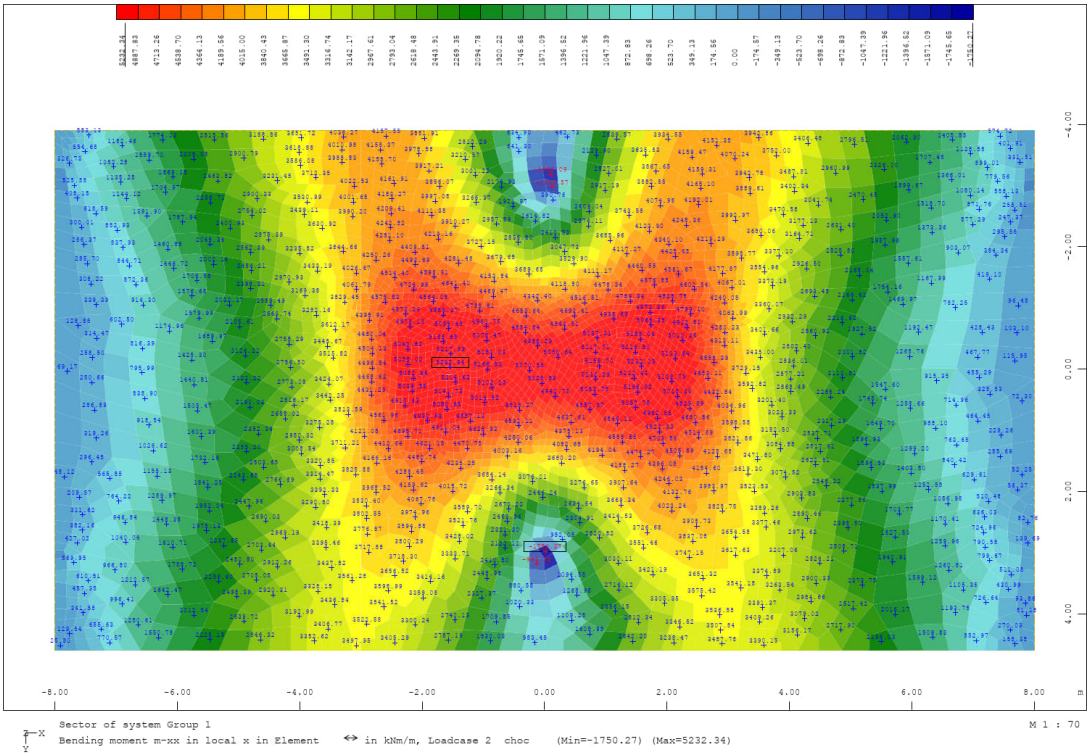


Figure 44. Bending moment distribution obtained from the static analysis.

The three bending moment distributions highlight the strong influence of the modelling approach on the internal force pattern.

In the linear static analysis, the bending moments are largely distributed over the central part of the slab, with a wide zone of very high positive moments. This reflects the direct

application of the peak load as an equivalent static pressure, which leads to a broad and highly conservative flexural response.

In the linear dynamic analysis, the moment field remains dominated by the central impact zone, but the distribution is more structured and less uniform than in the static case. The response is influenced by the transient nature of the loading and by the dynamic interaction between the slab and its supports, which limits the spatial extent of the highest moments.

In the nonlinear dynamic analysis, the maximum bending moments are significantly reduced and become more localised, with a clear redistribution of internal forces away from the centre. This behaviour is consistent with the development of nonlinear mechanisms such as cracking and stiffness degradation, which reduce the peak flexural demand and modify the load transfer within the slab.

Table 4 summarises the maximum vertical displacement and bending moment obtained for the different modelling approaches, although these maxima may occur at different locations in the structure.

Table 4. Comparison of maximum displacement and bending moment for the different analyses

Analysis type	u_z (mm)	m_{xx} (kNm/m)
Linear dynamic analysis	35.1	1520
Nonlinear dynamic analysis	98.6	643.8
Linear static analysis	166	5232
Nonlinear static analysis	No convergence	–

The comparison between the different modelling approaches highlights several important aspects of the structural response.

The most significant difference is observed between the static and dynamic analyses. Applying a static load corresponding to the peak dynamic force leads to substantially larger displacements and bending moments than those obtained in the dynamic simulations. This result clearly illustrates the overly conservative nature of representing the impact through an equivalent static load applied over the contact surface.

It should be noted that the dynamic amplification effect between static and dynamic responses cannot be observed in this case. In the dynamic simulations, the load is generated through a contact interaction between the falling container and the slab, allowing the force to develop progressively during the impact event. In contrast, the static analysis applies the peak force instantaneously as a uniform surface load over the contact area.

As a consequence, the static representation does not capture the transient interaction between the impacting mass and the structure, nor the redistribution of energy during the impact. The simplified static approach therefore introduces unrealistically high structural demands and leads to a significant overestimation of both internal forces and displacements.

This example clearly illustrates the limitations of using an equivalent static load to represent impact events. While such an approach is simple to implement, it does not adequately represent the physics of the problem and may lead to excessively conservative design assumptions.

The second important observation concerns the comparison between the linear and non-

linear dynamic analyses. The nonlinear dynamic analysis predicts substantially larger displacements than the linear dynamic model, indicating that additional deformation mechanisms are activated when material and geometric nonlinearities are taken into account.

At the same time, the nonlinear model predicts significantly lower bending moments than the linear dynamic analysis. This behaviour can be explained by the activation of energy dissipation mechanisms in the nonlinear model. Part of the impact energy is absorbed through concrete cracking, reinforcement yielding, and increased structural deformation, which reduces the forces transmitted through the structure.

From a structural design perspective, this observation is particularly relevant. While the nonlinear dynamic model predicts larger deformations, it results in lower internal forces compared to the linear dynamic analysis, which assumes purely elastic behaviour and therefore tends to overestimate the structural demand.

5 CONCLUSION AND FUTURE WORK

This thesis investigated the structural response of reinforced concrete elements subjected to blast type loading using both equivalent static approaches and dynamic time-history analyses. The work aimed to assess the relevance and potential advantages of dynamic modelling within an engineering design context using the SOFiSTiK finite element environment.

A consistent numerical framework was first developed to represent blast pressure-time histories based on the Kingery–Bulmash empirical correlations and the UFC 3-340-02 guidelines. The modelling approach was verified through comparison with published studies involving reinforced concrete beams and slabs subjected to blast loading. The results obtained with SOFiSTiK showed good agreement with the reference data, confirming that the implemented methodology provides a reliable representation of the structural response.

The proposed modelling approach was then applied to two engineering case studies. The first case concerned the blast assessment of a digester structure subjected to an internal gas deflagration. The comparison between equivalent static and nonlinear dynamic analyses highlighted the importance of considering the temporal evolution of pressure and the interaction between structural deformation and pressure relief mechanisms. The dynamic analysis predicted significantly larger displacements, illustrating the influence of transient effects during the explosion.

The second case study investigated the accidental drop of a heavy container on a reinforced concrete protection slab. The results showed that nonlinear dynamic analysis leads to larger displacements but lower internal forces compared with linear dynamic modelling. This behaviour can be explained by energy dissipation through material nonlinearities and structural deformation during the impact event.

From an engineering perspective, the results demonstrate that simplified static approaches may lead to overly conservative estimates when applied to highly dynamic phenomena such as blast loads or impacts. Nonlinear dynamic analysis provides a more realistic representation of structural behaviour and may allow a more rational structural design while maintaining appropriate safety levels.

Nevertheless, several limitations remain. In order to maintain a tractable modelling framework, certain simplifications were adopted in the present study. In particular, the structural model relies on shell-type QUAD elements to represent the reinforced concrete slab. While this approach allows efficient nonlinear dynamic simulations, a more detailed volumetric modelling using three-dimensional solid elements could provide a more refined representation of the stress distribution within the concrete and of the interaction between concrete and reinforcement.

In addition, the blast loading was represented through simplified pressure–time functions derived from empirical correlations. Although this approach is widely used in engineering practice, more detailed blast load descriptions could be investigated in future work in order to better capture complex explosion phenomena.

Future developments could also focus on extending the modelling framework to more complex structural systems and configurations. Further comparisons with experimental data would also be valuable to refine and validate the proposed methodology.

Overall, this work demonstrates that nonlinear dynamic finite element analysis represents a powerful tool for assessing reinforced concrete structures subjected to blast loading and provides a practical modelling strategy for engineering applications within SOFiSTiK.

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